Early embryo differentiation in zebrafish

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1 Glossary and preliminaries

Because biology likes to use big, fancy words, we first have to build a glossary of terms that will appear in this report. Most are relatively "standard" in the literature, but many have slightly different meaning across subfields, so this section can be also viewed as term definitions.

- A **transcription factor (TF)** is a protein that directly influences the transcription process, either by participating in it, or by modifying the environment that influences the actual transcription machinery (e.g., opening or closing chromatin). Note that this is intentionally broader than the strict textbook definition, which only covers proteins that bind specific DNA sequences; here, we also include co-factors and chromatin remodelers.
- Time points in zebrafish embryo development are usually denoted in **hours post fertilization (hpf)**. We will shorten this to **hours (h)** when clear from context. We will be mostly interested in the development processes taking place until approx. **12hpf**.
- Since zebrafish are simple organisms, the embryo development is also relatively stable and easy to reproduce. The most important environmental factor is **temperature**, which is typically **28.5 °C**. Other temperatures within acceptable rates should have roughly linear influence on the development speed [*citation needed*]. Other factors like pH, medium density or oxygen levels also impact development, but the embryo is robust to small, reasonable variations in them [*citation needed*].
- An embryo cell can be **totipotent**, **pluripotent**, or **multipotent**. Totipotent means it can differentiate into internal embryo cells, or external cell layers enveloping the embryo (**extra-embryonic cells**). Pluripotent means it can differentiate into any embryo cell, but not into an extra-embryonic cell. Multipotent means the cell can differentiate within a specific lineage (e.g., endoderm), but not outside of that specific lineage (at least not without a significant intervention).
- We can also distinguish cells by how *robustly* they are differentiated so far. Note that these terms are originally defined based on what happens to the cell across various environments, not necessarily by the epigenetics and transcriptome, so we are "extrapolating" a bit here to what is causing these terms to be well defined in the first place.
  - A cell is **primed** if its epigenetics admit a particular differentiation event (relevant chromatin is open, important TF is expressed but not causing transcription yet, etc.). A cell can be primed towards many lineages at the same time. Importantly, a primed cell will not differentiate without some additional "signal" triggering the differentiation.
  - A cell is **specified** if it is actively transcribing RNA of a particular cell type, and will continue this process in a neutral environment. In other words, the transcription has been started and is somewhat self-sustaining, but can be easily disrupted by an unexpected "signal".

  - A cell is **determined** if it is specified, but it can maintain its own identity when encountering an opposing, but reasonable "signal" (i.e., a signal that would naturally occur somewhere in the organism). Determination can be driven by a stronger self-sustaining GRN feedback loop, as well as by epigenetic changes.
  - A cell is **differentiated** if it is determined, and it has undergone morphological changes associated with its cell type, including additional epigenetic changes to further increase robustness against signaling perturbations.
  - The expression profile of a specified / determined / differentiated cell can be different, despite being "the same cell type". The cell is actively producing TFs that will lead to epigenetic changes required for determination / differentiation, but this transcription can eventually turn off once the cell is "sufficiently committed".
  - The process is hierarchical (cell type A differentiates into B, which turns into C, ...), and for the intermediate cell types, the cell does not have to pass all stages. For example, pluripotent cells will never be truly differentiated, or even determined. Similarly, stages can interleave across cell types, so a cell can be determined type A, but specified type B, and primed type C, meaning it will maintain identity A on its own, is already developing expression profile of type B, and is able to eventually become type C if a signal is given.

- A zebrafish embryo goes through several "high-level" developmental stages: **zygote (0-0.75h)**, **cleavage (0.75-2.25h)**, **blastula (2.25-5.25h)**, and **gastrula (5.25-10h)**.
  - In the zygote and cleavage stage, the embryo looks mostly uniform. However, it is not fully symmetric: some maternal factors (dorsal determinants, germ plasm) are already localized asymmetrically (see MBT below).
  - During blastula, the surface layer of extra-embryonic cells is starting to split off. These are the **enveloping layer** (EVL) and **yolk syncytial layer** (YSL); one guards the embryo from the environment, and one is an interface with the yolk on which the embryo rests. Remaining cells are also called **deep cells**.
  - A **midblastula transition (MBT)** occurs around **2.75hpf**. Until that point, the cells are predominantly driven by the maternal transcriptome present in the zygote (a small first wave of zygotic transcription already starts around the 64-cell stage). Until MBT, the whole cell population is essentially synchronous. Nevertheless, several important decisions are already made in this period, driven by maternal factors:
    * formation of the YSL and EVL (YSL forms at 512-1024 cells; Figure 1);
    * specification of a small group of **primordial germ cells**, which inherit maternal **germ plasm** during cleavage;
    * the **dorsal-ventral** axis (see below), determined by maternal dorsal determinants that are transported to one side of the embryo.

    The causes of these decisions mostly can't be observed through RNA expression (only the consequences), since they rely on localization of maternal proteins. Also, *the overall mass of the embryo does not change much* during blastula and gastrula (the cells mostly thin out and spread); actual growth comes later. After MBT, the cells divide asynchronously, hence there is no longer a clear cell doubling pattern.
  - During gastrula, the embryo splits along the *radial axis* (the deep–superficial direction, perpendicular to the embryo surface; not to be confused with the radial symmetry of the blastula around the animal-vegetal axis) into **hypoblast** (develops into **mesendoderm**) and **epiblast** (develops into **ectoderm**). Note that hypoblast and epiblast are purely positional terms: Both are part of the **germ ring**, with hypoblast being the inner layer (adjacent to the yolk), while epiblast is the outer layer (see also

Figure 1). In other words, a cell can be a part of hypoblast and of mesendoderm; one is the location, the other is the expression profile. The germ ring is the **margin** of the embryo (you can see cells called "marginal", meaning they belong to this region). The process of separation into hypoblast/epiblast along the germ ring is called **internalization**. During gastrula, the epiblast still contains mesendoderm precursors that internalize later; only once internalization finishes does epiblast fully correspond to ectoderm.

  - Within the gastrula phase, the embryo also visibly splits along the **dorsal-ventral** axis (dorsal is back / spine, ventral is front / belly). This split is marked by a **shield** formation on the dorsal side of the embryo. Note that the dorsal side itself is determined much earlier (maternally, see MBT above); the shield is where it first becomes morphologically visible. This is again mostly a positional term (i.e., shield is not an expression profile). However, cells in the shield do produce various signaling molecules that reinforce the dorsal-ventral split, and later also the anterior-posterior separation (see below). Sometimes, they are called **organizer cells**.
  - The term **lateral** (and its opposite, **medial**) is used in two related senses. During gastrula, it refers to position along the margin: viewed from the animal pole, the margin is a circle with the dorsal side (shield) on one end and the ventral side on the opposite end; the *lateral* margin are the two sides in between (left and right). In the body, it refers to the distance from the midline (the future spine): *medial* is close to the midline, *lateral* is far from it. The two are connected, because during gastrula the cells converge towards the dorsal side, which becomes the midline of the body. A related term is **paraxial**, meaning directly alongside the midline (i.e., the body axis).
  - After gastrula, the embryo undergoes **early segmentation**, where the split along the **anterior-posterior** (A-P) axis becomes morphologically visible (as with the other axes, A-P patterning, e.g., of the neural plate, already starts during gastrula). The A-P axis is often confused with the **animal-vegetal** axis, but these are different axes: animal-vegetal is the axis of the egg (animal pole = top, where the blastoderm sits; vegetal pole = bottom of the yolk), while A-P is the body axis, which only forms during gastrula. They are only roughly aligned (the animal region gives rise to anterior head structures, while the tail bud forms near the vegetal pole). In Figure 1, AP/VP mark the animal/vegetal pole, while Ant./Pos. mark anterior/posterior. Segmentation is beyond the scope of this document.

- During blastula and gastrula, the embryo also undergoes **epiboly**, during which it envelops the connected yolk. This is again largely used as a timing / position marker, with Figure 1 showing the individual stages. Note that the formation of a hypoblast and epiblast, as well as of a germ ring and shield are happening relatively quickly between 50% and 60% epiboly, but this is a morphological change; transcriptomic and epigenetic changes leading up to the formation are already happening sooner.

- Another note is that in terms of morphology and transcriptome, epiblast is essentially just the continuation of the existing deep cells, without major changes at the moment of separation. It is the hypoblast that represents a new lineage that is splitting off of epiblast in the marginal region. However, later during gastrula, the ectoderm itself splits into **neural** and **non-neural** ectoderm (the latter becoming the **epidermis**), driven by a signaling gradient along the dorsal-ventral axis, and this split is visible in the transcriptome.

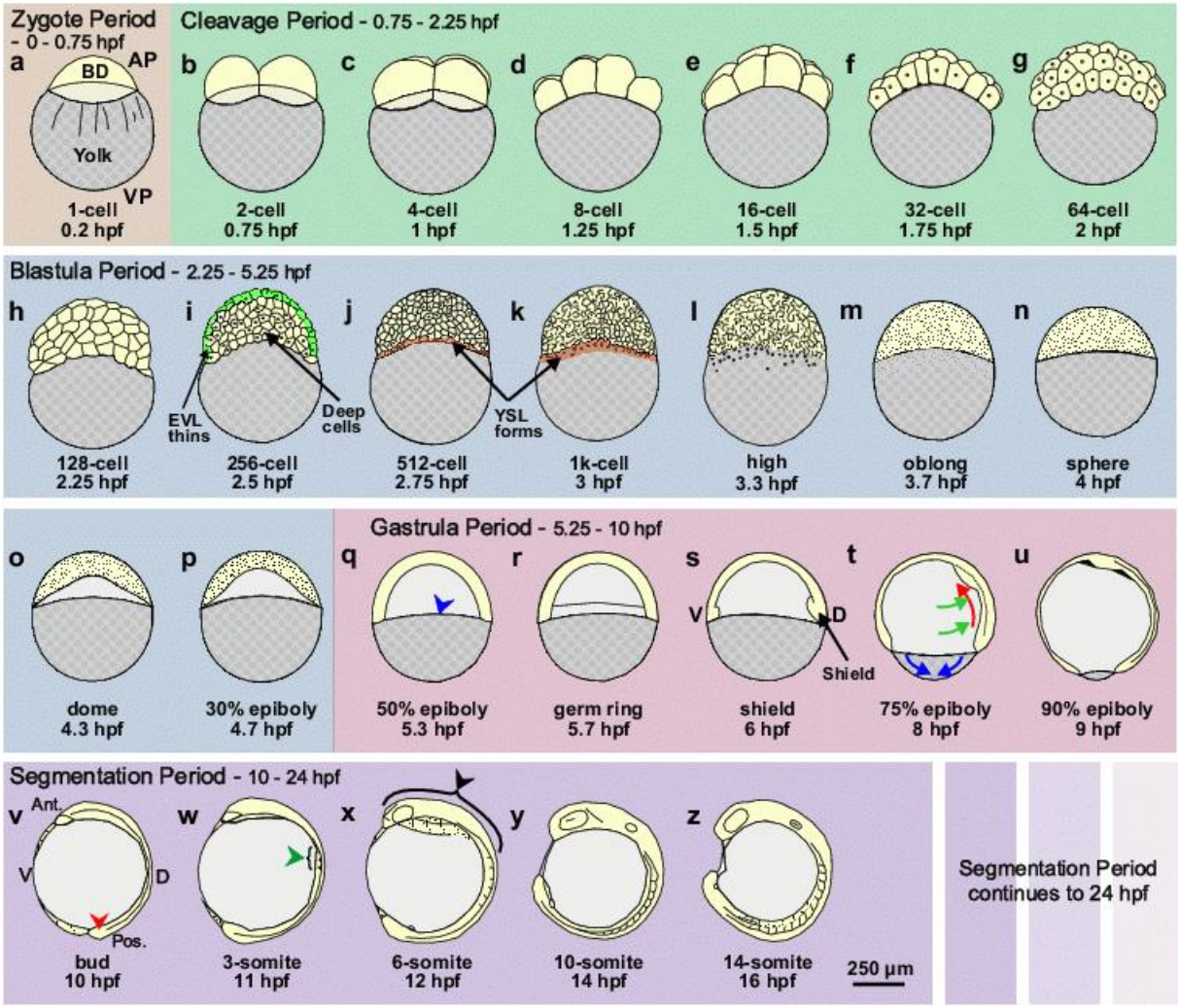


Figure 1: `https://www.aups.org.au/Proceedings/38/43-51/`

2 Transcriptomic and epigenetic timeline

In the previous section, we described the important terms and formations which appear during embryo development. However, these are mostly *morphological* terms dictating what the embryo looks like or what is happening externally. They do not capture how the cells are evolving internally, which typically precedes morphological changes quite a bit. Also, as we discussed, these do not map directly to any meaningful definition of "cell types" (i.e., a "shield cell" is not a well defined cell type). In this section, we thus summarize the *expression programs* that we expect to observe in the transcriptome during the first 12 hours, roughly when they emerge or diminish, what they do, and which earlier program they develop from. A few general remarks:

- All time ranges are approximate. They denote when a program should become *distinguishable in the transcriptome*, which usually precedes the corresponding morphological change by an hour or more. Also, the transitions are gradual: a new program typically first appears as a weak, graded bias within its parent population, and only later becomes a clearly separated cluster.
- Programs rarely "end"; most of them keep splitting into more specific sub-programs. When we say a program diminishes, we mean it is no longer a meaningful description of the cells, because they have been replaced by its more specific descendants.
- Not everything is equally visible in single-cell data. Some programs are not captured at all (e.g., the YSL, which is a syncytium, not a set of cells), or involve only a handful of cells (e.g., germ cells), and the GSE106587 atlas only starts at 3.3hpf, i.e., after MBT.

Table 1 gives an overview, and Figure 2 shows the same programs as a timeline, highlighting the branching points and the spatial axis along which each split happens. The individual programs are described in more detail below.

2.1 Cleavage and blastula (0-5.25hpf)

- **Maternal program** (0hpf, cleared by approx. 4-6hpf). Transcripts deposited in the egg by the mother. They drive the cleavage divisions and activate the zygotic genome. After MBT, a large fraction of maternal transcripts is actively degraded; by the start of gastrula, the transcriptome is dominated by zygotic expression [*citation needed*]. In the atlas, the earliest stages (high, oblong) should show a strong maternal component shared by all cells.
- **Blastomeres** (approx. 2-3hpf, diminishing 5-8hpf). The "default" program of deep cells after zygotic genome activation (ZGA). Cells are pluripotent and largely uniform; the observable differences between them are initially small and graded (distance from the margin, i.e., position along the animal-vegetal axis, and position along the dorsal-ventral axis). This program gradually disappears as the cells commit to germ layers; it persists longest in the epiblast, which is its direct continuation.
- **Yolk syncytial layer (YSL)** (2.75-3hpf, persists). Forms when marginal blastomeres collapse into the yolk cell. It drives epiboly and, importantly, is a source of signals that induce the marginal mesendoderm and contribute to the dorsal organizer [*citation needed*]. Because it is a syncytium inside the yolk, *it is not captured in single-cell data*; we only see its effect on neighbouring cells.
- **Enveloping layer (EVL)** (approx. 3-4hpf, persists as **periderm**). Develops from the outermost blastomeres, which become lineage-restricted during the blastula [*citation needed*]. Forms a protective epithelium around the embryo. In single-cell data, it is expected to be the first clearly separate cluster, already at the earliest atlas stages.

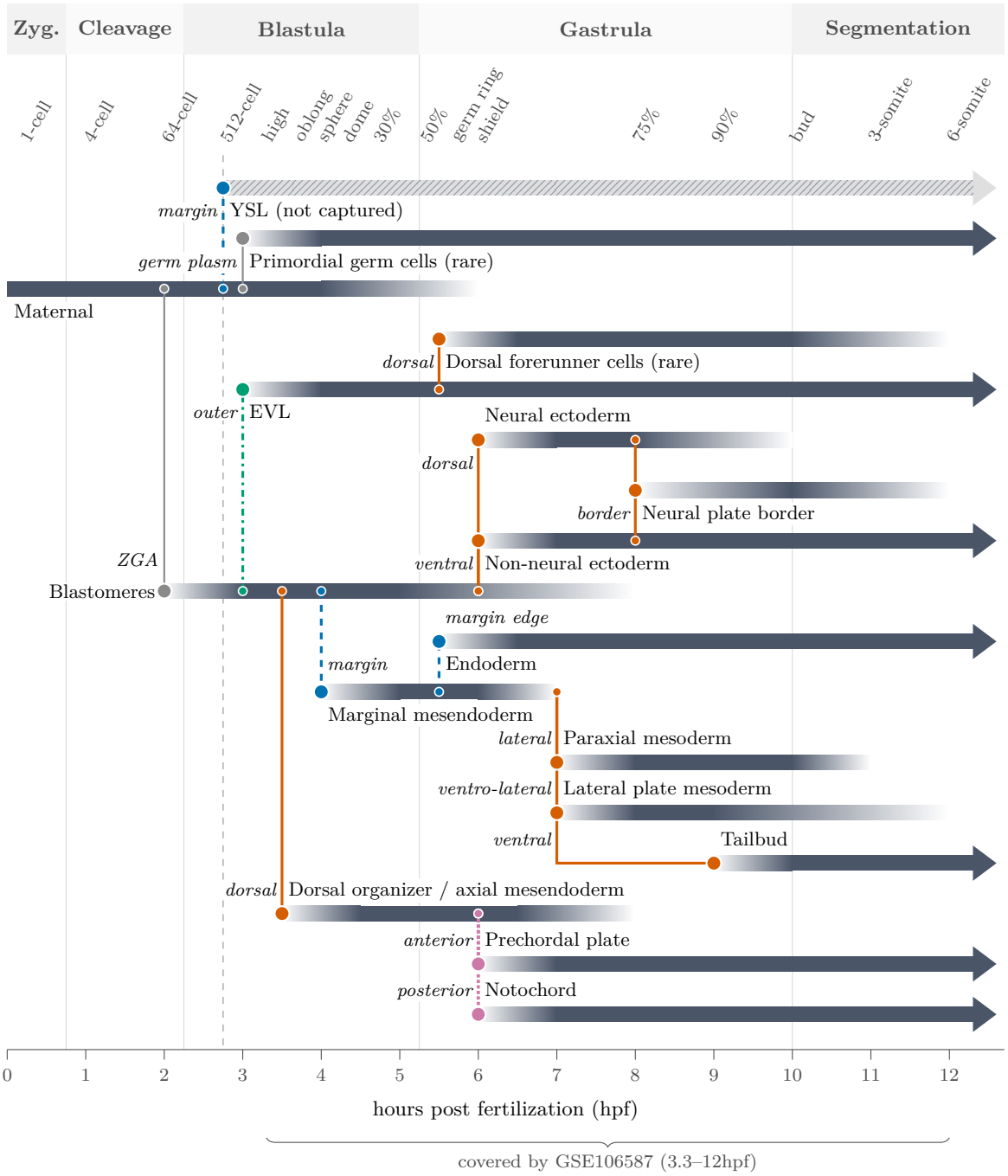


Figure 2: Timeline of the expected expression programs (Table 1). Each bar is one program; shaded ends mark the (approximate) intervals in which it emerges or diminishes. The dashed vertical line marks the MBT.

Program	Emerges (hpf)	Diminishes (hpf)	Develops from	Conditions	Note
Maternal	0	4-6	—	deposited in the egg	actively degraded after MBT
Blastomeres	2-3	5-8	maternal	zygotic genome activation (MBT)	replaced by germ layer programs; persists longest in the epiblast
YSL	2.75-3	—	maternal	marginal blastomeres fusing with the yolk	syncytium; not captured in single-cell data
EVL	3-4	—	blastomeres	outermost cell layer	becomes periderm
Primordial germ cells	3-4	—	maternal	inherited germ plasm	rare (a handful of cells)
Dorsal organizer	3.5-4.5	6.5-8	blastomeres	dorsal margin; maternal dorsal determinants	morphologically the shield (6hpf); splits into prechordal plate and notochord
Marginal mesendoderm	4-5	6-7	blastomeres	margin; signaling from YSL	forms the hypoblast; splits into endoderm and mesoderm programs
Endoderm	5.5-6.5	—	marginal mesendoderm	closest to the margin; strongest signaling	few cells at this stage; gut, liver, pancreas, pharynx
Dorsal forerunner cells	5.5-6.5	10-12	EVL	dorsal margin, ahead of the shield	rare; form the Kupffer's vesicle
Prechordal plate	6-7	—	dorsal organizer	first internalized cells	anterior part becomes the hatching gland (10-12hpf)
Notochord	6-7	—	dorsal organizer	later internalized cells	
Paraxial mesoderm	7-8	10	marginal mesendoderm	lateral margin; further from the margin edge	splits into adaxial cells and presomitic mesoderm; segments into somites
Lateral plate mesoderm	7-8	9-12	marginal mesendoderm	ventral and lateral margin	splits into heart, blood, endothelial and kidney precursors
Non-neural ectoderm	6-7	—	blastomeres	ventral epiblast; ventralizing signaling	becomes epidermis
Neural ectoderm	6-7	8-10	blastomeres	dorsal epiblast; protected by the dorsal organizer	subdivides along the A-P axis (forebrain, midbrain, hindbrain, spinal cord)
Neural plate border	8-10	10-12	neural and non-neural ectoderm	boundary between the two; intermediate signaling	splits into neural crest and placodes
Tailbud	9-10	—	marginal mesendoderm	ventral margin converging at the vegetal pole	includes overlying epiblast; neuromesodermal progenitors

Table 1: Overview of expected expression programs. Times are approximate and refer to when the program becomes distinguishable in the transcriptome; — means the program persists beyond 12hpf. *Develops from* is always a preceding program in the table; *conditions* describe where (location) or how (signaling, inheritance) the new program arises within it.

- **Primordial germ cells (PGCs)** (specified during cleavage, own zygotic program from approx. 3-4hpf, persists). Blastomeres that inherited maternal germ plasm. They give rise to gametes and do not participate in somatic differentiation. Only a handful of cells per embryo, so they will form a very small cluster.

- **Dorsal organizer** (approx. 3.5-4.5hpf, splits at 6.5-8hpf). Dorsal marginal cells (together with the dorsal YSL) that respond to the maternally determined dorsal side of the embryo. They produce signals that dorsalize the surrounding tissue, protect the dorsal ectoderm from ventralizing signals (enabling neural fate), and later contribute to A-P patterning. Morphologically, this becomes the **shield** at 6hpf; transcriptionally, the program is present well before that. Its cells later give rise to the axial mesoderm (prechordal plate and notochord). From approx. 6hpf, the internalized organizer cells (the inner layer of the shield) are often called the **axial mesendoderm**, i.e., midline mesendoderm that has not yet split into prechordal plate and notochord. We do not treat it as a separate program, because it is transcriptionally a gradual continuation of the organizer program; the clear split only happens with the prechordal plate and notochord.

- **Marginal mesendoderm** (approx. 4-5hpf, splits at 6-7hpf). Cells within a few cell tiers of the margin, induced by signals from the YSL and neighbouring marginal cells. The program is graded: the closer to the margin (and the stronger the signal), the more the cells are biased towards endoderm, while cells further away are biased towards mesoderm [*citation needed*]. These are the cells that internalize to form the hypoblast.

2.2 Gastrula (5.25-10hpf)

- **Endoderm** (approx. 5.5-6.5hpf, persists). Develops from the marginal mesendoderm cells that received the strongest inducing signal (mostly close to the margin, more on the dorsal side). Will form the gut and associated organs (liver, pancreas) and the pharynx. Relatively few cells at this stage.

- **Dorsal forerunner cells** (approx. 5.5-6.5hpf, forming the **Kupffer's vesicle** at approx. 10-12hpf). A small group of cells at the dorsal margin, ahead of the shield, partially derived from the EVL [*citation needed*]. They do not internalize; instead, they migrate ahead of the margin and form the Kupffer's vesicle, which establishes the left-right asymmetry. Only a handful of cells.

- **Prechordal plate** (approx. 6-7hpf, persists). The anterior part of the axial mesoderm, derived from the leading (first internalized) cells of the dorsal organizer. It migrates towards the animal pole and contributes to head patterning. Its most anterior part becomes the **hatching gland** (approx. 10-12hpf).

- **Notochord** (approx. 6-7hpf, persists). The posterior part of the axial mesoderm, derived from the remaining dorsal organizer cells. Forms the midline rod that later supports the body axis, and is an important source of signals that pattern the neighbouring neural tube and paraxial mesoderm.

- **Paraxial mesoderm** (approx. 7-8hpf, starts to form somites at 10hpf). Mesoderm from the lateral margin that converges on both sides of the notochord. Splits into **adaxial cells** (directly adjacent to the notochord; slow muscle precursors) and **presomitic mesoderm**, which will be segmented into somites (muscle, skeleton).

- **Lateral plate mesoderm** (approx. 7-8hpf, splitting at 9-12hpf). Mesoderm from the ventral and lateral margin. Gives rise to the heart, blood and endothelial precursors, and the pronephros (kidney). Many of these sub-programs only become clearly distinct around the end of gastrula or early segmentation.

- **Non-neural ectoderm** (approx. 6-7hpf, persists as **epidermis**). The ventral part of the epiblast, where ventralizing signals are strong. It is the first ectodermal program to diverge from the blastomere program.
- **Neural ectoderm** (approx. 6-7hpf, subdivided at 8-10hpf). The dorsal part of the epiblast, protected from ventralizing signals by the dorsal organizer. Forms the **neural plate**. During late gastrula, it is patterned along the A-P axis into forebrain (including eye field), midbrain, hindbrain and spinal cord programs.
- **Neural plate border** (approx. 8-10hpf, splitting at 10-12hpf). Cells at the boundary between neural and non-neural ectoderm, receiving intermediate levels of signals. Gives rise to the **neural crest** (very diverse lineage: pigment cells, craniofacial skeleton, peripheral neurons, ...) and **placodes** (sensory organs).
- **Tailbud** (approx. 9-10hpf, persists). Forms from ventral marginal cells converging at the vegetal pole at the end of epiboly. Contains bipotent **neuromesodermal progenitors** that produce both spinal cord and paraxial mesoderm during body axis elongation. Note that this is one of the few programs which mix "neural" and "mesodermal" signatures.

3 Dataset overview

The analysis is based on the GEO SuperSeries GSE106587 [7]. It consists of three single-cell datasets:

- **Drop-seq atlas** (SubSeries GSE106474). Wild-type embryos at 12 stages from high (3.3hpf) to 6-somite (12hpf); 39,505 cells, 23,974 genes. This is the main dataset.
- **SMART-seq2** (SubSeries GSE106466). 416 cells at 50% epiboly: 52 wild-type cells and 364 cells of MZ*oep* mutants, which lack Nodal signalling and hence most of the mesendoderm. Fewer cells, but deeper coverage per cell.
- **10x Genomics** (GSE106587 itself). Wild-type (3,000 barcodes) and MZ*oep* (2,200 barcodes) embryos at 6-somite (12hpf). The barcode lists are not QC-filtered, so not every barcode is a cell.

All three are available only as processed count matrices (raw reads were not downloaded). In the rest of this section, we only describe the Drop-seq atlas. **The GEO matrix has already passed the basic quality filters of the original study, but not its later outlier removal** (Section 3.4); here, we use all 39,505 cells without any additional filtering. The experimental details below are taken from the supplementary materials of the paper. The numbers, figures and table are produced by `scripts/01_dataset_structure.py`.

3.1 Stages, batches and libraries

The Drop-seq atlas has three nested levels (Figure 3 and Table 2):

- **Stage**: the developmental time point (12 stages, each with a single hpf value).
- **Sequencing batch** (DS2, DS3, DS4 and DS5): the "sequencing batch" field of the GEO samples. It is the main expected source of technical variation.
- **Library**: one sample of pooled embryos, processed in one Drop-seq run; equivalently, one GEO sample (GSM) and one SRA experiment. There are 28 libraries in total. Most (stage, batch) pairs have exactly one library. Six pairs have two, in which case the cell names of the second library carry a `b` suffix (e.g., `ZF50_WT_DS4b_...`).

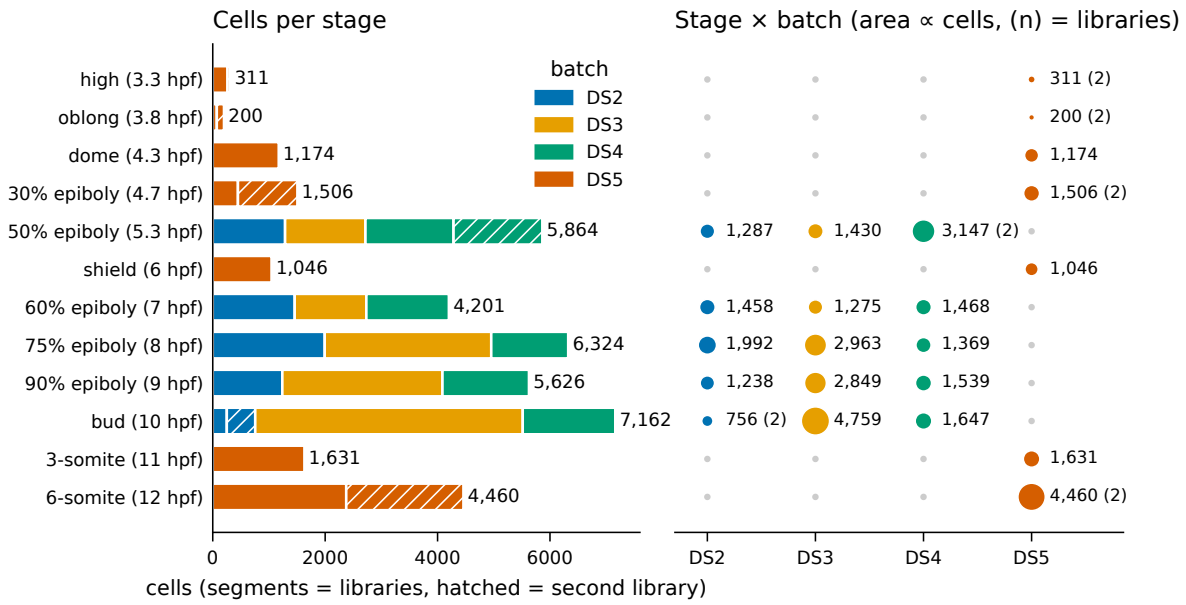


Figure 3: Structure of the Drop-seq atlas (GSE106474). Left: number of cells per stage; each segment is one library, coloured by sequencing batch, and second libraries of the same stage and batch are hatched. Right: the same counts per stage and batch. Stages of batch DS5 and stages of batches DS2–DS4 do not overlap.

Each library was prepared from a pool of 20–40 embryos of the same stage. The embryos were manually deyolked (**which also removes the YSL**), mechanically dissociated, and the resulting cell suspension was encapsulated in Drop-seq droplets. Consequently, cells carry no information about which embryo (or which position in the embryo) they came from. Some libraries were sequenced twice to increase depth, but the reads of such re-sequencing are merged into the same library; they do not form a second library.

Second libraries are separate embryo pools. GEO itself does not say whether the second library of a (stage, batch) pair comes from a separate embryo pool (biological replicate) or is a second run of the same pool (technical replicate). However, the supplementary Table S1 of the paper lists the total number of embryos used for each stage (Table S1). For high (56 embryos), oblong (76) and 30% epiboly (59), this is more than a single pool of at most 40 embryos, so the two libraries of these stages must come from different pools. For the remaining stages with a second library (50% epiboly, bud and 6-somite), the embryo counts are consistent with separate pools of approx. 20 embryos each, but they do not rule out a shared pool. The paper also treats each library as its own unit: Table S1 calls the libraries "batches" (two for high, four for 50% epiboly), and the two 6-somite libraries are described as "acquired in two batches" and batch-corrected before trajectory reconstruction. We thus treat second libraries as biological replicates sequenced in the same sequencing batch. Note that the paper's "batch" corresponds to our library, not to the GEO sequencing batch.

GEO replicate numbering. The "Rep N" in the GEO sample titles numbers the libraries of a stage *across batches*. It does not correspond to our a/b label. For example, "50% epiboly Rep 2" is the DS3 library, while "high Rep 2" is the second DS5 library of the high stage.

Table 2: Drop-seq libraries. Lib. "b" is the second library of the same stage and batch (a separate embryo pool). Medians are per cell, over all cells in the GEO matrix.

Stage	hpf	Batch	Lib.	GEO sample	Cells	Median UMIs	Median genes
high	3.3	DS5	a	GSM2838529	263	6,731	3,009
		DS5	b	GSM2838530	48	13,385	4,566
oblong	3.8	DS5	a	GSM2838531	72	11,866	4,171
		DS5	b	GSM2838532	128	3,603	2,048
dome	4.3	DS5	a	GSM2838528	1,174	4,398	1,931
30% epiboly	4.7	DS5	a	GSM2838506	448	3,858	1,717
		DS5	b	GSM2838507	1,058	3,260	1,564
50% epiboly	5.3	DS2	a	GSM2838509	1,287	5,600	2,106
		DS3	a	GSM2838510	1,430	4,562	1,730
		DS4	a	GSM2838511	1,567	5,179	1,988
		DS4	b	GSM2838512	1,580	5,630	1,962
shield	6	DS5	a	GSM2838533	1,046	2,904	1,291
60% epiboly	7	DS2	a	GSM2838513	1,458	6,173	1,898
		DS3	a	GSM2838514	1,275	6,074	1,826
		DS4	a	GSM2838515	1,468	4,970	1,640
75% epiboly	8	DS2	a	GSM2838518	1,992	5,404	1,768
		DS3	a	GSM2838519	2,963	3,352	1,258
		DS4	a	GSM2838520	1,369	5,582	1,712
90% epiboly	9	DS2	a	GSM2838521	1,238	5,814	1,793
		DS3	a	GSM2838522	2,849	3,377	1,282
		DS4	a	GSM2838523	1,539	3,603	1,343
bud	10	DS2	a	GSM2838524	250	6,642	1,915
		DS2	b	GSM2838525	506	6,446	1,864
		DS3	a	GSM2838526	4,759	2,556	1,063
		DS4	a	GSM2838527	1,647	3,683	1,381
3-somite	11	DS5	a	GSM2838508	1,631	2,609	1,047
6-somite	12	DS5	a	GSM2838516	2,375	2,000	884
		DS5	b	GSM2838517	2,085	3,180	1,237

3.2 Batch design

The most important property of the atlas is that the batches split the stages into two blocks that *share no stage*:

- **Block A** (batch DS5 only): high, oblong, dome, 30% epiboly, shield, 3-somite and 6-somite; 11 libraries, 10,328 cells.
- **Block B** (batches DS2, DS3 and DS4): 50% epiboly, 60%, 75% and 90% epiboly, and bud; 17 libraries, 29,177 cells. Every stage of this block was sequenced in all three batches.

Within block B, the design is balanced, so stage and batch effects can be separated. Within block A, there is only one batch, so stage differences are not confounded with batch, but there is also nothing to estimate the DS5 batch effect from. This is particularly unfortunate for our focus on the decisions around 6hpf. Shield (6hpf, DS5) lies between 50% epiboly (5.3hpf) and 60% epiboly (7hpf), which are both in block B. A batch effect will therefore appear as an abrupt developmental change at shield and back again at 60% epiboly.

3.3 Library size and sequencing depth

The number of cells per stage ranges from 200 (oblong) to 7,162 (bud) (Figure 3). The five block B stages (50% epiboly to bud) contain 74% of all cells. The two earliest stages are tiny: high and oblong have 511 cells together, and three of their four libraries have fewer than 150 cells (48, 72 and 128). Sequencing depth (Figure 4) is a median of 3,798 UMIs and 1,444 detected genes per cell over the whole atlas. We can make the following observations:

- **Depth decreases with developmental time**, from library medians of 4–13 thousand UMIs per cell at high and oblong to 2–3 thousand at 3- and 6-somite. Part of this is biological (cells get smaller and contain less mRNA with each division, which is also why the original study used stage-specific filtering thresholds).
- **Libraries of the same stage and batch are not interchangeable.** At oblong, the median depth of the first library is 3.3× that of the second; at high, the second library is 2× deeper than the first, and at 6-somite 1.6×.

3.4 Cell filtering

The original study filtered the cells in two steps; the dataset includes the first, but not the second filtering.

Basic quality filters (already applied in GEO). First, cells were filtered by the number of detected genes and UMIs, using a separate range for each stage (Table S1), because RNA content per cell decreases over early development. Cells below the range are likely low-quality (e.g., broken cells or empty droplets with ambient RNA), while cells above the range are likely doublets. Afterwards, cells with more than 45% mitochondrial RNA (a sign of stressed or dying cells) were removed. The GEO matrix is consistent with these filters having been applied:

- Only 51 of the 39,505 cells fall outside the gene ranges, all of them just slightly.
- The mean number of detected genes per stage agrees with Table S1 to within 1.5%.
- The GEO matrix contains no mitochondrial genes. This explains why the mean UMIs per cell are 20–25% lower than in Table S1 (where approx. 20–29% of UMIs are mitochondrial), and why some cells fall below the UMI ranges.

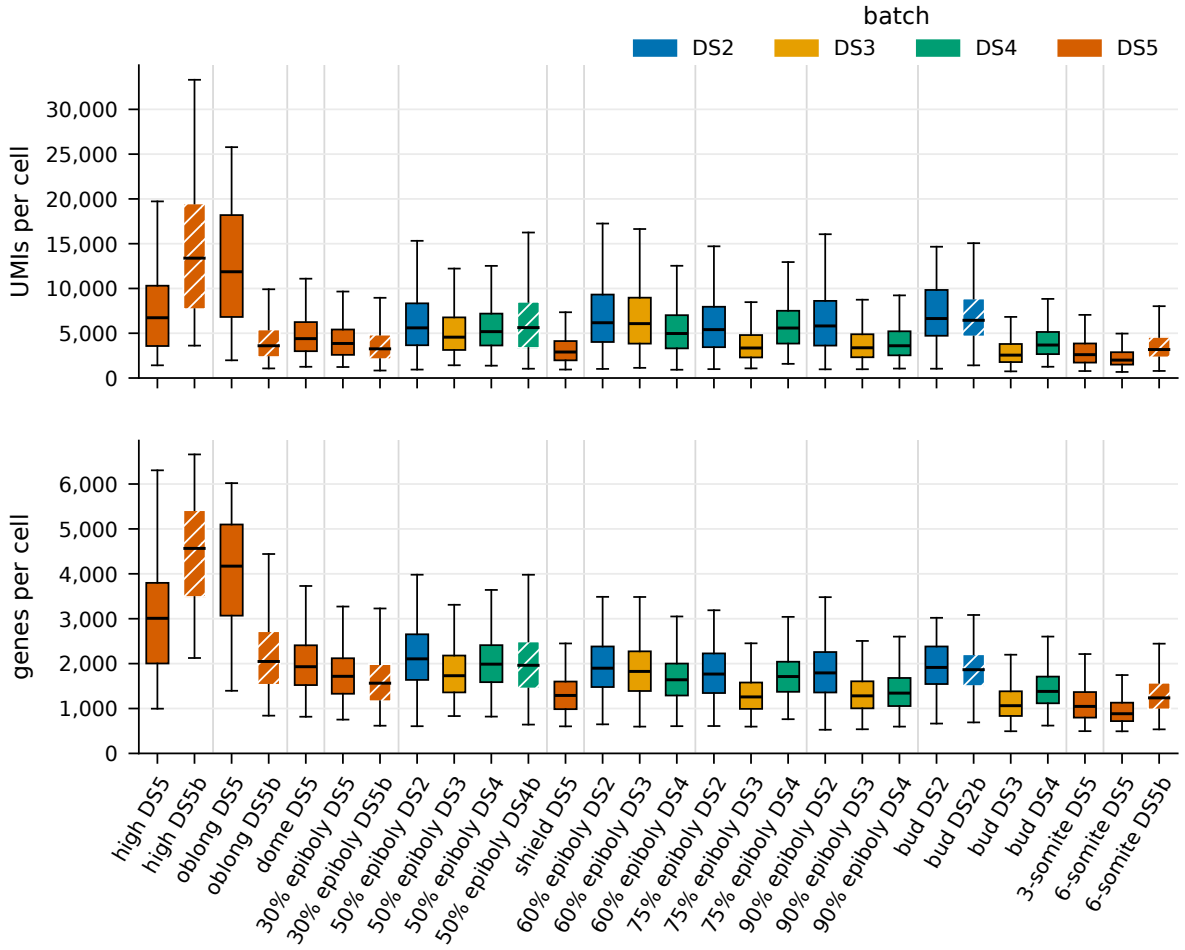


Figure 4: Per-library sequencing depth: UMIs (top) and detected genes (bottom) per cell. Boxes show the quartiles, whiskers extend to 1.5× the interquartile range (outliers not shown). Colour is the sequencing batch; second libraries are hatched. Thin vertical lines separate stages.

Outlier removal (not applied in GEO). Before the trajectory reconstruction, the paper removed 618 cells forming a cluster dominated by an "apoptotic-like" expression program, which appears in cells of many different types and would therefore create artificial connections between unrelated cell types. It also removed 139 cells that were poorly connected in the nearest-neighbour graph. The paper's full-data visualization uses 39,488 cells, and $39{,}488 - 618 - 139 = 38{,}731$, which is the number of cells reported in the paper. The GEO matrix contains 17 cells more than 39,488, which the paper does not explain. The identities of the removed cells are not available in GEO. Finally, the per-stage cell counts in Table S1 sum to 38,671, i.e., 60 fewer than 38,731, so the paper is not fully consistent here either.

3.5 Consequences for the analysis

- **Quantitative analysis within library, qualitative comparison across libraries.** To avoid batch and stage effects, quantitative analysis (normalization, binarization, etc.) should be calibrated and applied within individual libraries. Between libraries, we should primarily draw conclusions from qualitative results (e.g., gene A is higher than B in cells X across DS2/DS3/DS4; the quantity can be different, but the effect stays the same).

- **When to compare quantities across libraries?** We will most likely need to break the

first rule in a few places. First, comparing **time stages within batch** is reasonable when assessing long-term changes in transcription (but should probably carry greater burden of proof than within-library tests). Similarly, **batch-effect calibration within stage** is possible for DS2, DS3 and DS4. This is not possible with respect to DS5, but maybe we can at least extrapolate some useful information from it. Batch correction ideally stays within each stage to avoid "correcting for" biological development. But most algorithms can take this into account, so in theory, for DS2/DS3/DS4, a full batch correction run is viable; however, it is questionable whether it is needed to derive useful qualitative comparisons. Finally, **library-effect within batch and stage** can be also quantified using the a/b replicates, but we only have these for a small number of libraries, and they do come from separate embryo pools, so there can be genuine biological differences between them.

- **Some pseudo-libraries may be needed.** The early development stages (high, oblong) contain a very small number of cells, further subdivided into the a/b replicates. Assuming we don't observe significant expression profile differences between the a/b replicates, it may be reasonable to merge them and treat them as one library to obtain a bigger cell pool for the following experiments. In this context, **down-sampling or weighting the remaining, much larger libraries may be necessary** for some analysis methods to avoid over-representation of the 50%/60%/75%/90%/bud stages.

- **Outliers must be handled by us.** The GEO matrix still contains the apoptotic-like cells and the poorly connected outliers removed in the paper. We likely need our own (comparable) outlier filtering step. Furthermore, some cell types not relevant for the Boolean model may be also eliminated outright (e.g. the germ cells, or the EVL). However, for operations such as discretization, it may be useful to have the whole expression distribution, even if it is not fully assigned to cell types.

4 Result: markers of heterogeneity within libraries

Many of the following analyses need a set of "interesting" genes: genes that define a low-dimensional embedding of the cells, genes whose expression we try to predict, or genes that pass a filter because they mark a subpopulation. The standard choice is a fixed number of *highly variable genes* (HVGs). In this section, we replace it with a test that asks, for every gene and library, whether the gene varies between the cells of the library more than a gene expressed equally in every cell would. Intuitively, we expect genes that define specific subpopulations of the library to fall within such a subset. **We call such genes *heterogeneous*** (within a library).

The test relies on the assumption that the **UMIs of a uniformly expressed gene should follow the negative-binomial distribution**, which accounts for the variance of bursty gene expression (the telegraph model) and sampling noise [14, 34]. The test quantifies the distance of the real expression from such an ideal distribution. The cutoff is calibrated against simulated libraries of uniformly expressed genes, which leaves approx. 10 false positives per library, and combined with an effect size estimated from the library itself, which gives three tiers of genes with an estimated false discovery rate of 50%, 10% and 1% (weak, moderate, strong). **The test selects 421–1,527 genes per stage** (weak tier, dome onward; high and oblong are mostly uniform with 33–164 het. genes per library). We show that the test is more consistent across comparable libraries than HVGs and it aligns better with genes with a localized expression pattern according to ZFIN [1] records; the advantage is large over the original Seurat HVGs and moderate over newer HVG methods. The numbers, figures and tables are produced by `scripts/01b_heterogeneous_genes.py`, `scripts/01c_heterogeneity_zfin.py` and `scripts/01d_hvg_flavours.py`.

4.1 A model of a homogeneous gene

Consider one library and one gene. Cell i has a relative depth s_i (its number of UMIs divided by the library average); in this dataset, s_i varies by approx. 10× within a library. The variation in s_i is mostly technical (capture efficiency), but it also absorbs differences in cell size: we only assume that a homogeneous gene makes up the same fraction of every cell's transcripts. Therefore, if the gene were to be expressed at the same level in every cell, cell i would contain on average $s_i \cdot m$ of its UMIs, where m is the gene's mean per average cell. However, the observed count of such a homogeneous gene is still random for two reasons:

- **Sampling.** Only a fraction of the transcripts of a cell is captured and sequenced. Given the true expression, the captured count follows Poisson distribution, with variance $s_i \cdot m$.
- **Biological noise.** Even in identical cells, the true expression is not identical: transcription occurs in bursts [34], and cells can differ in size and cell-cycle stage. If the true expression varies between cells with a squared coefficient of variation φ (gamma-distributed), the count is negative binomial (NB), with variance $s_i \cdot m + \varphi \cdot (s_i \cdot m)^2$.

NB counts describe droplet-based UMI data well, without any need for extra "dropout" zeros [14, 39, 15]. Meanwhile, the Poisson model ($\varphi = 0$) is not a useful baseline: For a well-expressed gene, the extra variance $\varphi \cdot (s_i \cdot m)^2$ is small in each cell, but with a thousand cells it is easily detectable: at shield, 88% of the genes with 100–200 UMIs have $\hat{\varphi} > 0$. Under the Poisson model, nearly every well-expressed gene would have higher variance than expected. In other words, if we only accounted for sampling noise (Poisson model), but not for the biological expression noise (negative binomial model), we would wrongly conclude most genes are heterogeneous.

However, in the NB model, we do not know the true biological value of φ for each gene. Instead, we estimate its value based on *typical genes with the same mean*, and then ask whether the observed φ differs significantly from this "typical" φ_t (in other words, we assume most genes are homogeneous, which allows us to estimate their φ_t and use it to distinguish the heterogeneous ones). A gene restricted to a subpopulation has a much larger φ: if it is expressed in a fraction f of the cells (and uniformly within them), $\varphi \approx 1/f - 1$, e.g., 9 for $f = 10\%$ (this is only the impact of the subpopulation; true φ will also depend on the biological noise; consequently, φ is not zero even if $f = 100\%$).

Estimating the squared coefficient of variation. We estimate φ separately for every gene in every library. Let y_i be the UMI count of the gene in cell i, where i runs over all cells of the library (no cells are removed). Then we use the method of moments:

$$\hat{\varphi} = \frac{\sum_i (y_i - s_i\hat{m})^2 - \sum_i s_i\hat{m}}{\hat{m}^2 \sum_i s_i^2}, \qquad \hat{m} = \frac{\sum_i y_i}{\sum_i s_i}.$$

Here $\hat{m}$ is the gene's mean per average cell, the first sum is the observed variance around the expected counts $s_i\hat{m}$, and the second sum is the part of it that Poisson sampling explains. Thus $\hat{\varphi}$ is the variance left after subtracting the sampling noise, divided by the scale at which φ enters the variance. The estimate is 0 when the gene varies exactly as much as sampling alone predicts, and negative when it varies less, which happens by chance.

We only estimate φ for genes with at least 10 UMIs in the library (6,756–12,555 of the 23,974 genes, depending on the library). Below that, a gene is detected in a handful of cells and its variance says almost nothing about its distribution. Such a gene is considered *not measured* in that library: it has no estimate and is never considered heterogeneous there. In particular, it also counts as not heterogeneous when we combine multiple libraries of a single stage (see below).

The *typical* φ_t at a given mean is the value of a dispersion–mean trend fitted to the genes of the same library (see next paragraph). The *residual* of a gene is $\log_{10}\hat{\varphi} - \log_{10}\varphi_t$, i.e., $\log_{10}$

of the ratio of the two: 0 means typical, 1 means ten times more variable than typical. We use a ratio rather than a difference because φ_t itself changes by orders of magnitude with the mean. An absolute difference of 0.5 would thus be negligible for rare genes but extreme for well-expressed ones. Meanwhile, ratio remains comparable across genes with vastly different φ_t. The ratio requires a positive $\hat{\varphi}$, so negative and zero estimates are floored at 10^{-3}; such genes are far below any reasonable cutoff and are clearly not heterogeneous anyway.

Typical overdispersion: a dispersion–mean trend. In every library, genes of similar mean have similar $\hat{\varphi}$, and $\hat{\varphi}$ falls with the mean. We describe this by a trend with three parameters,

$$\varphi_t(m) = a_0 + a_1\, m^{-b}.$$

Its shape follows from the mechanism of the noise. Under the telegraph model of bursty transcription [34], a gene's NB shape parameter is its burst frequency and its mean is the burst frequency times the burst size, so $\varphi \approx$ burst size$/m$: rarer genes are noisier, approx. as $1/m$ if burst sizes are similar. The constant a_0 is the noise floor shared by all genes (e.g., cell-cycle stage and cell size beyond what s_i absorbs), which becomes comparable to the power law only for the most expressed genes; the same "a/mean + floor" pattern was found across the proteome of single *E. coli* cells [41]. Comparing every gene with this trend thus absorbs the variation that all genes share, without having to model each source.

The trend is fitted only to genes with at least 100 UMIs in the library. Below that, the estimate is dominated by chance: a gene with 10–30 UMIs in a library of 1,000 cells is detected in a few dozen cells, typically with one UMI each. For example, at shield, only 33% of the genes with 10–15 UMIs have a positive estimate, compared with 88% of the genes with 100–200 UMIs. For genes with fewer than 100 UMIs, $\varphi_t(m)$ is therefore a pure extrapolation. To check that the extrapolation does not depend on this choice, we refitted the trend to genes with at least 50 and 30 UMIs instead of 100: the typical value of the rarer genes changes by a median of at most 0.01 in $\log_{10}$ (at most 0.11, in the smallest library from dome on). At 20 UMIs, the fit breaks in some libraries, because the bins of such genes have medians at or below 0, i.e., the threshold must stay well above the point where most estimates are dominated by sampling.

The genes with at least 100 UMIs are sorted by mean and divided into bins of 200 genes, and $\log_{10}\varphi_t$ is fitted by least squares to the medians of $\hat{\varphi}$ in the bins (6–35 bins per library). The medians make the fit insensitive to heterogeneous genes, which are a minority of every bin. In the 24 libraries from dome on, the trend describes the bin medians within 0.03 in $\log_{10}$ (root mean square), with very similar exponents $b = 0.72$–0.85 (median 0.77), i.e., slightly less steep than $1/m$, as expected if burst sizes grow somewhat with expression. The floor is small ($a_0 = 0$–0.026, median 0.008): over the observed range, the trend is mostly the power law, and it levels off only for the most expressed genes. There, it is slightly too high in the late libraries (median residual of the 100 most expressed, non-heterogeneous genes -0.15 to -0.22 at 3- and 6-somite, -0.11 to $+0.07$ elsewhere), which makes the test more conservative for these genes. In the four libraries of high and oblong (48–263 cells), only 6–20 bins are available and their medians are close to the noise of the estimate; the fitted curves are flat or erratic ($b = 0$–0.66), the typical value of rare genes is unreliable (Figure S1), and which of their genes are selected depends mostly on the chance level and on high effect-size floors (see below).

Relation to DESeq2 and other methods. The model is the one used for bulk RNA-seq by DESeq2 [27]: NB counts with variance $\mu + \alpha\mu^2$, size factors that play the role of s_i, and a dispersion α (our φ) that is assumed to depend smoothly on the mean, with the parametric trend $\alpha(\mu) = a_0 + a_1/\mu$ (ours has a free exponent). The two methods use the trend in opposite ways. DESeq2 has few samples per condition, so each gene's dispersion is very uncertain; the trend serves as a prior towards which the gene-wise estimates are shrunk, so that the *means* of the conditions can be compared reliably. Deviations from the trend are treated as noise,

except for "dispersion outliers" (genes far above the trend), which are not shrunk because they plausibly have extra variability. We have hundreds to thousands of cells per library, so the gene-wise estimates are precise for well-expressed genes, and the deviations from the trend are exactly what we look for: our heterogeneous genes are the dispersion outliers of a single library. In single-cell data, sctransform [15] also regularizes NB parameters as a smooth function of the mean, but uses them mainly to normalize cells; the variance of its residuals is used to rank genes, without a calibrated cutoff; and Brennecke et al. [2] select genes whose CV^2 lies above a trend of the same form, $a_0 + a_1/\mu$, in which, unlike in our φ, the sampling noise is included.

Selecting heterogeneous genes. A residual above 0 does not yet make a gene heterogeneous. A homogeneous gene can lie above the trend for two reasons:

1. Its estimate is noisy (*sampling noise*), which matters most for rare genes; this is handled by a **chance level** obtained from simulated libraries.

2. It is intrinsically noisier than the median gene of its expression level (*intrinsic noise*, e.g., larger bursts), which matters most for well-expressed genes, where sampling noise is small; this is handled by an **effect-size floor** obtained from the spread of the real genes.

A gene is heterogeneous in a library if its residual reaches the larger of the two.

Sampling noise: chance level. Even relative to a sensible typical value, the residual of a rare gene is noisy. A fixed cutoff therefore does not work: a residual of at least 0.5 (3.2× typical) selects 953–1,977 genes in the libraries of more than 1,000 cells, but also 126–615 genes in a simulated library with the same depths s_i and gene means $\hat{m}$, in which every gene is drawn from an NB with the typical φ of its mean, i.e., in which no gene is heterogeneous; in the small high and oblong libraries, the simulation even reaches 54–78% of the genes selected in the data (Figure S2a). We therefore use such simulations to set the cutoff. Eight simulated copies of each library give, for every expression level, the 99.9% quantile of the residual of a homogeneous gene (running quantile over the approx. 1,000 genes of most similar mean, i.e., 8,000 simulated values; the residual is taken against the typical φ the gene was simulated with, so that the trend of φ with the mean does not enter the quantile). This chance level rises smoothly towards rare genes: at shield, from 2.4× typical above 1,000 UMIs to 13× at 10–15 UMIs. The simulations are random; each library has its own fixed seed (derived from its name), so the results are reproducible, and with other seeds, 2–4% of the genes selected in a library change (up to 15% in the smallest libraries; tested on four libraries).

Intrinsic noise: effect-size floor. The simulation gives every homogeneous gene exactly the typical φ, but real genes differ in their intrinsic noise even when they are expressed uniformly. Among well-expressed genes (at least 1,000 UMIs), the residuals of the bulk of the genes have a robust standard deviation of 0.26–0.38 (from dome on), approx. 3–4× more than in the simulation, while the chance level there is only 1.4–2.7× typical. The chance level alone would thus select every gene that is somewhat burstier than the median gene. We therefore require a minimum effect size, derived from an *empirical null* [6]: the residuals of the well-expressed genes (or of the 300 most expressed genes, if fewer are well expressed) are modelled as normal, with their median and robust standard deviation. For any floor, the estimated false discovery rate (FDR) is the number of well-expressed genes this null distribution would place above the cutoff, divided by the number of well-expressed genes that actually are above it. Each library gets three floors, the smallest ones with an estimated FDR of at most 50% (**weak**), 10% (**moderate**) and 1% (**strong**); a gene belongs to a tier if its residual reaches the larger of the tier's floor and the chance level. In the four libraries of high and oblong, there are hardly any well-expressed genes, and the 300 most expressed ones still carry much sampling noise (standard deviation 0.35–0.64), so their floors are high (0.83–1.49 for weak), and at oblong, the three tiers coincide.

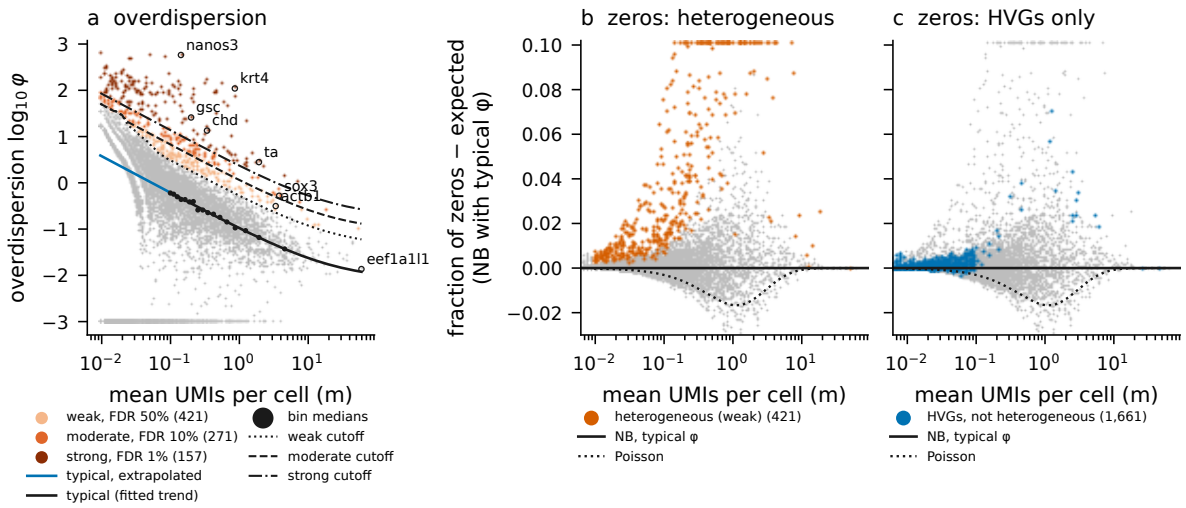


Figure 5: Heterogeneous genes in the shield library. (a) Estimated overdispersion against the mean (both logarithmic). Solid: typical value (fitted trend $a_0+a_1m^{-b}$), fitted to the bin medians (dots); blue where it is extrapolated to rarer genes. Dotted, dashed and dash-dotted: the cutoffs of the weak, moderate and strong tiers; light to dark orange: genes of these tiers. Estimates ≤ 0 are shown at -3. (b, c) Fraction of cells without a UMI of the gene, minus the fraction expected for a typical gene of the same mean (NB with typical φ, the library's depths), against the mean (logarithmic); values above 0.1 are drawn at the top of the axis. Dotted: the same for a Poisson gene. (b) heterogeneous genes (weak tier); (c) the HVGs of the library (Seurat, top 2,000) that are not heterogeneous. Grey: all other genes.

What the tiers mean. Table S2 summarizes the tiers and the two kinds of false positives. The sampling-noise false positives are counted in an independent, ninth simulation; the simulation cannot produce intrinsically noisy genes, because all its genes have the typical φ. The FDR, in turn, refers only to well-expressed genes, where the empirical null can be measured: only 17% (median) of the weakly heterogeneous genes are well expressed, so the intrinsically noisy ones are at most approx. 9% of the weak tier. Since the bulk of well-expressed genes also contains genes with mild, real differences between cells (e.g., gradients), its spread, and hence the FDR, is if anything overestimated. For rarer genes, intrinsic noise cannot be measured under the sampling noise; there, the chance level is already 4–13× typical, at or above two standard deviations of the intrinsic spread among well-expressed genes (approx. 4×), so it also excludes most of them.

Defaults. **Heterogeneous** without qualification means the weak tier. We use it by default, i.e., we accept that up to half of the well-expressed heterogeneous genes may be merely noisier than typical, in exchange for missing few real markers; the moderate and strong tiers are available where a smaller, more reliable set is needed. Rare genes are selected if their variation is large: at shield, *sox17* (endoderm) has only 26 UMIs, but a residual of 1.20 (16× typical), and is moderately heterogeneous. A gene is in a tier **at a stage** if it is in that tier in at least half of the stage's libraries (Table 3). The overdispersion, typical value and cutoffs of every library are shown in Figure S1.

Zeros. The same model also predicts how many cells should contain no UMI of a gene. For a cell with expected count $\mu_i = s_i m$, this probability is $(1+\varphi\mu_i)^{-1/\varphi}$ under the NB, and $e^{-\mu_i}$ under Poisson. A gene expressed in only some cells has *more* zeros than a typical gene of the same mean, because its UMIs are concentrated in fewer cells. Figure 5b shows these excess zeros at shield. Typical genes lie on the NB curve (0), and slightly above the Poisson curve,

which predicts too few zeros. The heterogeneous genes have up to 40% more cells without a UMI than expected (the figure is cut at 10%), i.e., they are absent from a substantial part of the library. The excess grows with the mean, as an expressed gene becomes increasingly unlikely to be missed by chance; at the lowest means, zeros are expected anyway, so the zero fraction carries little information, and the variance (panel a) is the more sensitive measure. Excess zeros are a consequence of heterogeneity, not a criterion: a gene can be heterogeneous with few extra zeros, if its variance comes from graded expression levels or from a few cells with many UMIs (36 of the 421 heterogeneous genes at shield have fewer extra zeros than an NB gene exactly at the weak cutoff would).

4.2 Why not highly variable genes?

The original Seurat HVG method [37] (Seurat v1/v2, still the default of scanpy [47]) normalizes every cell to 10,000 UMIs, takes $\log(1 + x)$, and ranks the genes by their dispersion (variance over mean) relative to genes in the same bin of mean expression, keeping the top 2,000. This has three problems:

- **Rare genes look variable.** After depth normalization and the logarithm, a gene with a single UMI in a few cells has a large variance relative to its tiny mean, and its bin contains only similarly rare genes. As a consequence, many HVGs are genes we cannot measure at all: per stage, 1,083–1,924 of the 2,000 HVGs are not heterogeneous, with a median of 5–22 UMIs in the *whole stage*, and 761–1,074 of them (38–54% of all HVGs) have fewer than 10 UMIs in every library (Table 3). Accordingly, the HVGs that are not heterogeneous have no more zeros than a typical gene: at shield, only 13 of these 1,661 genes have more than 2% extra zeros (median 0%; Figure 5c), against a median of 2.2% for the heterogeneous genes. Most of them are barely detected (median 0.005 UMIs per cell).
- **Depth is only removed from the mean.** Normalizing to a common total changes the scale, but not the sampling noise, which is larger in shallow cells; the log transform distorts it further [44]. The NB model accounts for depth explicitly.
- **Pooled libraries.** Selected on the pooled cells of a stage, a gene that differs between libraries (e.g., the heat shock genes of 50% epiboly DS4*a*, Section 5) is "variable" even if it is uniform within each library. Our test runs within each library.

Conversely, 82–610 heterogeneous genes per stage (from dome on) are not among the HVGs, although they are well expressed (median 120–1,578 UMIs in the stage). Newer HVG methods fix the first two problems in part; we compare with them in Section 4.4.

Replication. If a gene ranking reflects the cell types of a stage, it should be similar in two libraries of the same stage. Over all pairs of libraries of the same stage, the 1,000 genes with the highest residual overlap approx. twice as much (Jaccard index 0.27–0.50 from 30% epiboly on) as the 1,000 genes with the highest Seurat dispersion (0.14–0.29; Figure S2c). The same holds for the sets we actually use: the weakly heterogeneous genes of two libraries overlap 0.38–0.61, against 0.16–0.28 for as many top Seurat genes. At high and oblong, neither replicates (0.06–0.21).

4.3 Heterogeneous genes reflect known expression patterns

A gene selection that replicates better across libraries than HVGs could still be the wrong one, e.g., it could favour genes with some technical property. As an independent check, we ask whether genes known to be expressed in a restricted part of the embryo at a given stage are heterogeneous at that stage, and whether genes known to be expressed everywhere are not. The knowledge comes from the expression records of the Zebrafish Information Network (ZFIN) [1],

Table 3: Heterogeneous genes per stage. Weak, Mod., Strong: genes in the tier in at least half of the stage's libraries. The last three columns compare the heterogeneous (weak) genes with the 2,000 Seurat HVGs of the pooled stage (Both: in both sets); in brackets, the median number of UMIs of the genes in the whole stage.

Stage	Lib.	Cells	Weak	Mod.	Strong	Both	HVG only	Het. only
high	2	311	182	106	58	138	1,862 (22)	44 (62)
oblong	2	200	102	102	102	76	1,924 (14)	26 (37)
dome	1	1,174	429	258	175	347	1,653 (6)	82 (241)
30% epiboly	2	1,506	527	368	276	343	1,657 (7)	184 (340)
50% epiboly	4	5,864	823	602	453	570	1,430 (7)	253 (1329)
shield	1	1,046	421	271	157	339	1,661 (5)	82 (120)
60% epiboly	3	4,201	744	483	316	547	1,453 (9)	197 (1578)
75% epiboly	3	6,324	908	656	449	614	1,386 (8)	294 (1494)
90% epiboly	3	5,626	1,031	771	500	688	1,312 (8)	343 (949)
bud	4	7,162	953	676	422	617	1,383 (8)	336 (1405)
3-somite	1	1,631	899	608	301	612	1,388 (6)	287 (191)
6-somite	2	4,460	1,527	1,037	560	917	1,083 (5)	610 (264)

compiled from the literature and from large in situ screens. The numbers and table are produced by `scripts/01c_heterogeneity_zfin.py`.

ZFIN records. We use the wild-type expression records of ZFIN (downloaded on 2026-10-26; `scripts/download_zfin.sh`). Each record states that a gene was detected in an anatomical structure (e.g., "shield", "neural plate", "whole organism") during a range of developmental stages, by a given assay, in a given publication. We keep:

- **mRNA in situ hybridization** only. This is the assay that shows *where* in the embryo a gene is expressed; other assays (e.g., RT-PCR) record the whole embryo by necessity.
- **Publications before 2018**, so that the records are independent of the atlas [7]. The year is taken from ZFIN's publication list or, for direct data submissions such as the in situ screens of Thisse et al., from the date in the publication identifier.
- **Genes of the atlas**: ZFIN genes are matched to the gene symbols of the count matrix via their current symbol or an unambiguous former symbol (the matrix often uses older names).

This leaves 76,143 records of 8,350 genes (of 84,705 in situ records of 10,247 genes from 4,263 publications).

Stages. ZFIN stages are time intervals (e.g., "Shield" is 6–8hpf), and a record covers the time from the start of its first stage to the start of the stage after its last one. A record is assigned to an atlas stage if it covers the time of the atlas stage (the start of the ZFIN stage of the same name; 60% epiboly, which ZFIN does not distinguish, at 7hpf, i.e., within "Shield"). At each atlas stage, a gene is

- **restricted** if at least one of its records at that stage names a localised structure, i.e., any anatomy term except 15 broad terms ("whole organism", "unspecified", "ectoderm", "head", "trunk", ...; `data/curated/zfin_anatomy_programs.tsv`) and the yolk and YSL, which were removed with the yolk before dissociation;
- **whole organism** if all of its records at that stage name broad terms only, i.e., it was seen everywhere (or no more specific location was recorded).

Table 4: Heterogeneous genes and ZFIN expression records (in situ hybridization, before 2018) per stage. All measured: genes with at least 10 UMIs in one library of the stage, and the percentage of them that are heterogeneous. ZFIN: measurable genes that are restricted (recorded in a localised structure at that stage) or recorded in the whole organism only, and the percentage of them selected as heterogeneous (Het., weak tier), among the same number of top Seurat HVGs (HVG_k), or among the 2,000 top HVGs. High and oblong have too few restricted genes to be informative.

	All measured		ZFIN: restricted				ZFIN: whole organism only			
Stage	Genes	Het. %	Genes	Het. %	HVG_k %	HVG_{2000} %	Genes	Het. %	HVG_k %	HVG_{2000} %
high	9,515	2	15	47	27	53	3,105	1	1	9
oblong	8,062	1	10	40	10	50	2,887	0	1	10
dome	10,730	4	52	44	33	48	3,244	2	1	5
30% epiboly	9,978	5	131	61	40	57	3,204	2	1	5
50% epiboly	12,489	7	533	47	26	38	3,785	3	2	3
shield	8,370	5	552	35	18	36	3,295	1	1	6
60% epiboly	11,394	7	662	45	27	42	3,759	3	1	4
75% epiboly	11,847	8	652	51	29	44	3,729	4	1	4
90% epiboly	11,832	9	608	57	34	45	3,681	4	2	4
bud	12,623	8	813	48	27	40	3,840	4	1	3
3-somite	10,425	9	1,491	32	16	28	3,328	3	2	4
6-somite	11,992	13	1,763	40	23	27	3,567	4	2	3

We expect restricted genes to be heterogeneous, and whole-organism genes not. Neither expectation is strict: a restricted gene may be too rare to reach our cutoff, or expressed in a large domain (e.g., the whole neural plate), and a record may cover several stages of which only some show the restricted pattern. We compare three selections of the genes measurable at the stage (at least 10 UMIs in one of its libraries): the heterogeneous genes, the same number of top Seurat HVGs (HVG_k), and the usual 2,000 HVGs.

Results. Table 4 lists all stages. High and oblong are not informative: ZFIN records the blastula almost only as "whole organism", so there are only 15 and 10 measurable restricted genes, and these stages have few heterogeneous genes (Section 4); we exclude them from the summary. From dome to 6-somite:

- **Restricted genes are heterogeneous 3–12× more often than an average gene** (32–61% of them, median 46%, while the heterogeneous genes are 4–13% of the measured genes). In the moderate and strong tiers, 22–54% (median 35%) and 11–39% (median 26%) of them are selected.
- **The HVGs recover fewer restricted genes:** 16–40% (median 27%) for the same number of HVGs, and 28–57% (median 41%) for 2,000 HVGs, i.e., with 1.3–4.8× as many genes.
- **Whole-organism genes are depleted:** 1.4–4.4% of them are heterogeneous (weak), 0.3–0.5× the rate of an average gene, and 0.8–2.4% and 0.4–1.3% in the moderate and strong tiers. The method therefore does not simply select expressed or well-studied genes.

The recall is highest during gastrulation (30% epiboly: 61%; 90% epiboly: 57%) and lowest at 3- and 6-somite (32–40%), where ZFIN records many genes with organ-specific patterns that begin during segmentation and are recorded over long stage ranges (i.e., partially out of scope for the studied atlas).

4.4 Comparison with newer HVG methods

The test is a variant of an established idea: compare the variance of a gene with that of genes of similar mean under a count model, as do the technical-noise model of Brennecke et al. [2], the current Seurat default ("vst": the variance of the counts standardized by a fitted mean–variance trend [38]) and the analytic Pearson residuals [24] (the variance of NB Pearson residuals with one fixed dispersion). The comparisons of the previous sections use the oldest method, so we repeat them for the two newer ones, each on the pooled cells of a stage, and also with the genes ranked within each library and then combined (the recommended use with several batches). As usual, these methods first drop genes detected in fewer than 3 cells; we also show the old method with this filter. Figure 6 summarizes the stages from dome on (ranges in Table S3) (`scripts/01d_hvg_flavours.py`):

- **Most of the gap is the old method.** Of the ZFIN genes with a restricted pattern at a stage (Section 4.3), the top k genes of the old method contain 16–40%, those of both newer methods 25–56% and our ranking by the NB residual 29–59% (k = the number of heterogeneous genes of the stage). In replication, the top 1,000 genes of two libraries overlap 0.14–0.29 (old), 0.29–0.49 (vst), 0.24–0.46 (Pearson residuals) and 0.29–0.50 (NB residual; median pair per stage). The newer methods are close to our ranking: ours is ahead at most stages by at most 0.05, and behind at a few (e.g., Pearson residuals replicate better at bud). The filter of rare genes explains only a small part of the difference.

- **Rare genes remain.** Even the newer methods put 380–790 genes with fewer than 10 UMIs in every library of the stage among their top 2,000 when selected on the pooled stage (ranked per library, fewer at some stages, e.g., 1 at bud for Pearson residuals); in the top k, still 90–330 (pooled) or 0–256 (per library). Our test does not rank such genes at all.

- **The calibrated set is better than any ranking of the same size.** The heterogeneous genes contain 32–61% of the restricted genes, 2–9 percentage points more than the best ranking at every stage, also at single-library stages (shield 35% vs. 27–32%, 3-somite 32% vs. 25–29%), where it cannot profit from replication. The cutoff that depends on the mean (the chance level of Section 4) admits genes that a single ranking puts too low. And the calibration gives the number of genes (421–1,527 per stage), where an HVG method needs a fixed number such as 2,000.

We therefore keep the heterogeneous genes, but the advantage over modern HVG methods is moderate; most of what Table 3 shows is a weakness of the old method.

4.5 Consequences for the analysis

- **The heterogeneous genes of a stage replace the HVGs** wherever we need "interesting" genes: for embeddings and neighbourhoods, as evaluation targets, and as gene filters. By default, these are the weakly heterogeneous genes; where a smaller, more reliable set is needed (e.g., as markers of a population), the moderate or strong tier should be used. They are listed per stage and library, with their tier, in `results/tables/heterogeneous_genes.tsv.gz` and `..._calls.tsv.gz` (0: none, 1: weak, 2: moderate, 3: strong). The function `heterogeneity_calls` in the script applies the same calibration to any subset of cells (e.g., half of a library).

- **The two earliest stages have little detectable structure.** High and oblong have 182 and 102 heterogeneous genes, with 3–8 false positives per library: with 48–263 cells per library, only strong differences can be told from chance, and the rankings of their two libraries hardly overlap. This agrees with the expectation that cells before dome are largely equivalent (apart from small populations such as the germ plasm-carrying cells:

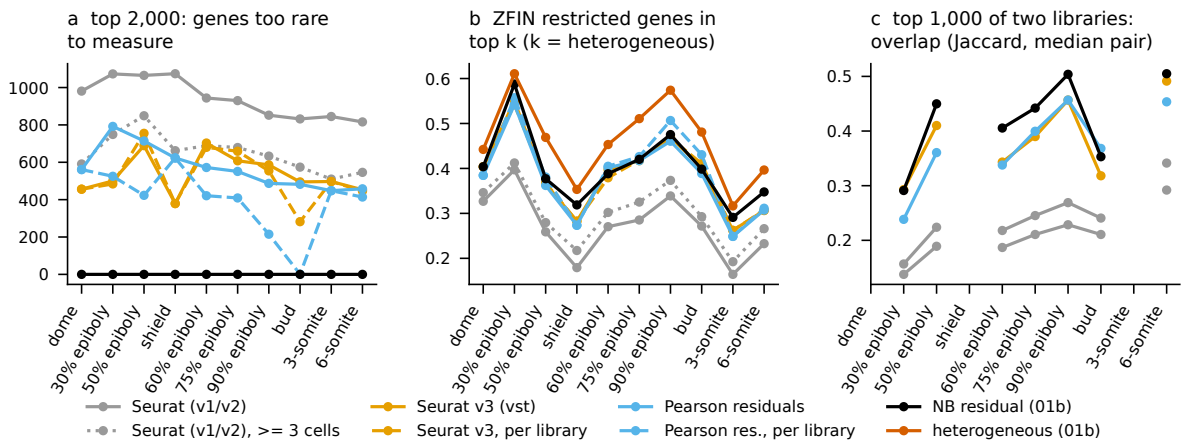


Figure 6: Heterogeneous genes vs. newer HVG methods, from dome on. (a) Genes with fewer than 10 UMIs in every library of the stage among the top 2,000 genes of each method (selected on the pooled stage, or per library and combined; dashed). (b) Share of the ZFIN genes with a restricted pattern at the stage among the top k genes, k = the number of heterogeneous genes. (c) Overlap of the top 1,000 genes of two libraries of the same stage (median over pairs; stages with two or more libraries).

nanos3 is strongly heterogeneous at every stage where it is measured), but it also means that we have little power there.

- **The number of heterogeneous genes grows over development,** from 421–1,031 between dome and bud to 1,527 at 6-somite, as cell types diversify. Shield (421) and dome (429) are single libraries; at stages with several libraries, a gene must be heterogeneous in at least half of them.

5 Result: a/b libraries have distinct expression profiles

Six (stage, batch) pairs have two libraries, a and b (Table 2). If the two libraries of a pair were interchangeable, we could merge them, which would remove one level of the library hierarchy and, in particular, give larger cell pools for the tiny high and oblong stages. In this section, we test whether the two libraries of a pair contain the same population of expression profiles. **They do not: in every pair, the libraries can be told apart, in some pairs as clearly as two neighbouring stages, and neither depth normalization nor depth matching changes this.** We therefore keep the a/b libraries separate. The numbers and figures are produced by `scripts/02_replicate_mixing.py` (mixing) and `scripts/03_replicate_differences.py` (gene-level evidence).

5.1 Experiment

For each pair, we embed the cells of both libraries together using a standard pipeline: each cell is normalized to 10,000 UMIs and log-transformed, the genes are restricted to those heterogeneous in either of the two libraries (Section 4; 74–1,527 genes per comparison), and the cells are projected onto the first 30 principal components using PCA. We do not use the usual highly variable genes: selected on two libraries jointly, they include genes that differ between the libraries as a whole rather than between cells, which would bias the test towards separation. In this space, we find the 20 nearest neighbours of every cell. If the libraries contain the same population of cells, the neighbourhood of a cell is a random sample of both libraries, i.e., the number of neighbours from library b follows a binomial distribution around the overall share of

b cells. If one library contains cells that the other lacks, or if all cells of one library are shifted in expression, the neighbourhoods are dominated by one library. We summarize this with three measures; the first two are standard metrics of batch mixing in single-cell data [28]:

- **Rejection rate** (in the style of kBET [3]): the fraction of cells whose neighbourhood composition differs from the overall composition (binomial test, $p < 0.05$).
- **Local inverse Simpson's index** (LISI [22]): the effective number of libraries in a neighbourhood (between 1 and 2). It is computed as $1/(p_a^2 + p_b^2)$, where p_a and p_b are the fractions of neighbours from each library, so a 50/50 neighbourhood gives 2 and a single-library neighbourhood gives 1. For simplicity, we weight the 20 neighbours uniformly instead of using the original distance-based weights.
- **Classifier AUC**: a logistic regression predicts the library of a cell from its principal components (5-fold cross-validation). This is a classifier two-sample test [26], a general statistical method rather than an established single-cell metric. A related idea was used by the authors of the atlas, who identified batch-specific gene modules by how well they predict the batch (ROC AUC, via Seurat's `BatchGene` function) [7]. An AUC of 0.5 means the libraries are indistinguishable, 1.0 means they are fully separable.

None of these has a natural threshold for "mixed well", and with thousands of cells, even a negligible difference becomes statistically significant. We therefore calibrate every pair against reference comparisons of the same size:

- **Permutation null**: the a/b labels are randomly shuffled (200 distinct repetitions) on the same neighbourhood graph. This is what perfect mixing looks like.
- **Negative control**: a random split of one library into two parts of the same sizes as the pair, analysed with the whole pipeline. This is what two samples of the same embryo pool look like.
- **Positive control**: library a against an adjacent stage of the same batch (e.g., 50% epiboly DS4 against 60% epiboly DS4). This is what a real developmental difference looks like.
- **Other batch** (where available): library a against the same stage from another batch (50% epiboly DS3, bud DS3).

Finally, we know that the libraries of a pair differ in sequencing depth (up to 3.3× in the median, Figure 4). Normalizing each cell to the same total removes differences in the average expression level, but not in sparsity: a shallow cell has more zeros, which can by itself move it in the embedding. We thus run every comparison twice:

- **Normalized**: all UMIs, only the per-cell normalization described above.
- **Thinned**: before normalization, each UMI of the deeper library is kept with probability equal to the ratio of the two median depths (binomial thinning [10]). Afterwards, both libraries have the same median depth, as if the deeper one had been sequenced less.

5.2 Results

The results are summarized in Figure 7. The full per-cell distributions of neighbourhood composition for all comparisons are in Figure S3, the UMAPs of all six pairs in Figure S4 (after thinning: `results/figures/replicate_mixing_umap_thinned.pdf`), and all numbers in `results/tables/replicate_mixing.tsv`.

- **The pipeline itself does not separate the libraries.** The random splits are indistinguishable from the permutation null (AUC 0.46–0.54, rejection rate at most 4%). AUCs slightly below 0.5 are not a systematic effect, but the noise of cross-validation, which goes in either direction.
- **All six pairs are clearly distinguishable.** The classifier AUC is 0.87–1.00, while the highest AUC in the permutation null is 0.52–0.65. Between 12% and 82% of cells have a neighbourhood dominated by one library, compared to at most 4% under the null.
- **The difference is sometimes comparable to a stage step.** At high, *a* against *b* is as separable as *a* against the adjacent stage (AUC 0.97 vs. 0.96). At 50% epiboly, DS4*a* is as different from DS4*b* as from DS3 (82% vs. 79% rejected neighbourhoods). The bud DS2 pair is the most similar (AUC 0.94, 12% rejected), but there, the libraries of different batches (DS2 against DS3) are also unusually similar (AUC 0.90, 24% rejected; after thinning even less separable than the pair, AUC 0.77 vs. 0.94).
- **Depth is not the cause.** In the normalized experiment, depth alone predicts the library with AUC up to 0.87 (oblong, where library *a* is 3.3× deeper; 0.76 at high). After thinning, this drops to 0.48–0.55 in all pairs, but the libraries remain separable (AUC 0.83–1.00).
- **The difference is spread over many genes.** Removing the 1,000 genes with the largest a/b fold change and repeating the analysis leaves five of the six pairs clearly separable (AUC 0.68–0.78, against at most 0.59 in the permutation null). Only the high pair becomes nearly indistinguishable (AUC 0.60, null up to 0.56).
- **The libraries share their structure, but are shifted within it.** In the UMAPs (Figure 7, bottom), both libraries cover the same branches, but within each branch, the cells of *a* and *b* occupy neighbouring, mostly non-overlapping regions. This suggests that most of the difference is a shift of the whole library rather than a cell population unique to one of them.

Consequently, merging the a/b libraries would mix two distinct expression profiles. Normalization or thinning does not make them compatible. If we need larger cell pools for some analysis, the two libraries have to be treated like two batches (i.e., with the same care as any other between-library comparison), not as one population. This agrees with the original study, which treated libraries as batches: at 50% epiboly (four libraries), 7 of the 20 gene modules it identified reflected batch effects and were removed, and the two 6-somite libraries were batch-corrected before trajectory reconstruction [7].

5.3 Hypotheses: why do the libraries differ?

The analysis above shows *that* the libraries differ, not *why*. Here, we collect the evidence for three plausible explanations. This is not meant to be exhaustive or conclusive. It is based on a pseudobulk comparison of each pair (mean expression per 10,000 UMIs, and the fraction of cells in which a gene is detected, computed after thinning the deeper library so that detection rates are comparable). With only one library per side, there is no replicate-level variance, so we do not report *p*-values. Table S4 lists the genes supporting each explanation; all genes with at least a 2-fold difference are in `results/tables/replicate_de_genes.tsv`.

Different parents (clutch effect). Each library is a separate pool of embryos, quite possibly from different parents. The supplement of [7] only states that the embryos come from wild-type crosses of the Tupfel longfin/AB background, a mix of two laboratory strains, and does not say whether the pools of one stage come from different crosses or days. Wild-type zebrafish lines are genetically heterogeneous, and before and shortly after zygotic genome activation, nearly

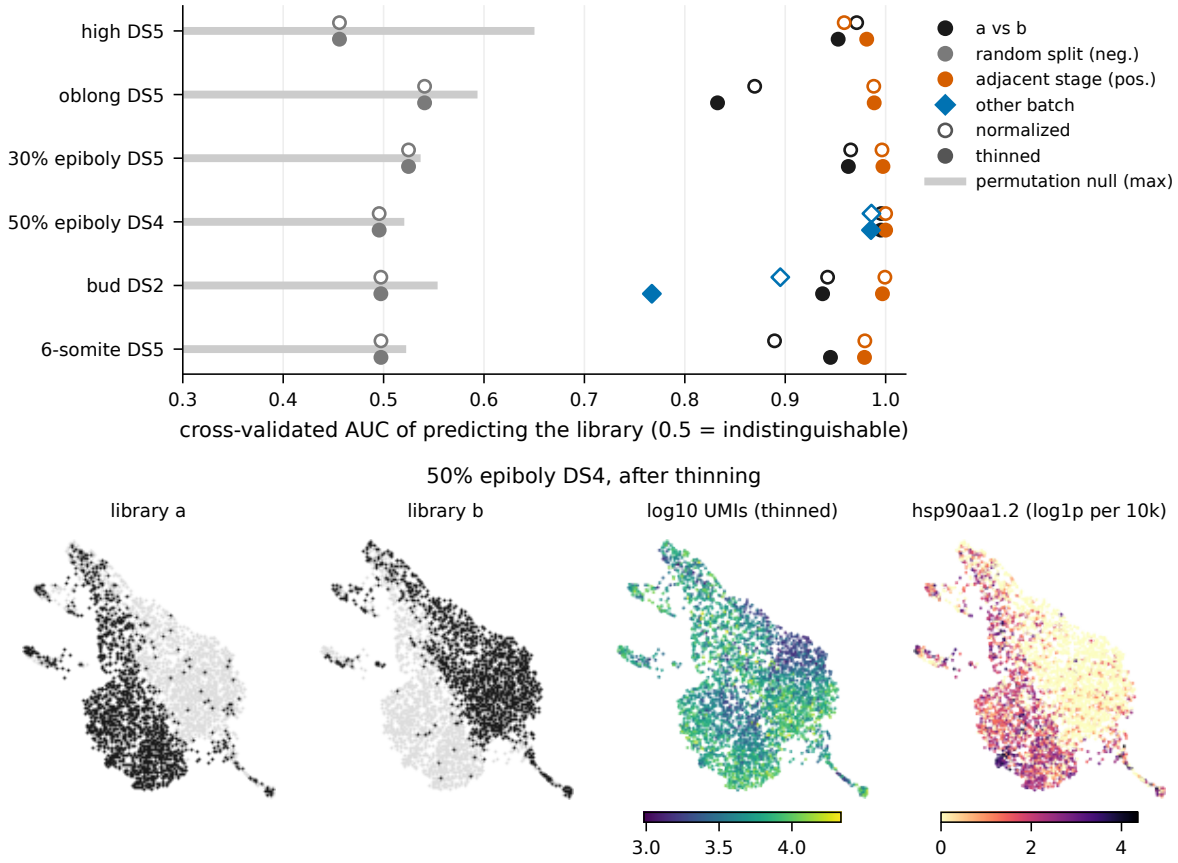


Figure 7: Mixing of the a/b libraries. Top: cross-validated AUC of a classifier predicting the library of a cell from its principal components, for each pair (a vs. b, black) and its reference comparisons; open markers without, filled markers with binomial thinning of the deeper library. The grey bar extends to the highest AUC obtained with randomly permuted labels. Bottom: UMAP of the thinned 50% epiboly DS4 pair, with separate panels for the cells of each library, the depth after thinning, and the expression of the heat shock gene *hsp90aa1.2* (Section 5.3).

all mRNA in a cell is inherited from the mother. A genetic difference between parents should show up as *on/off genes*, detected in many cells of one library and almost none of the other, regardless of cell type. Such genes exist in high (e.g., *gbgt1l4*: 6% vs. 71% of cells) and 50% epiboly DS4 (e.g., *si:dkey-108k21.10*: 49% vs. 5%), see Table 5. At high, library b also contains approx. 2–4× more of the maternal germ plasm transcripts *nanos3* and *dnd1*, which is consistent with a different maternal contribution. Checking whether the on/off genes lie in genomic regions known for copy-number variation would strengthen this explanation.

Handling stress. In the 50% epiboly DS4 pair, library a strongly expresses heat shock genes (*hsp90aa1.2* 16×, *hsp70.3* 9×, *serpinh1b* 7×). *hsp90aa1.2* is detected in 81% of the cells of library a, but only in 14% of library b, and it is high throughout library a in the UMAP (Figure 7), i.e., this is a shift of the whole library, not a subpopulation. Heat shock genes respond within minutes to hours, so this most likely reflects the handling of the embryos or cells (temperature, dissociation) rather than development. The protocol leaves room for such differences: dissecting one sample took up to 15 minutes, cells were dissociated mechanically, and droplets were held on ice for up to 1.5 hours to allow several sequential collections before reverse transcription [7], so two libraries of the same stage and batch may have been handled for different lengths of time. Note that this also affects any comparison involving DS4a, e.g.,

heat shock genes appear among the top differences between 50% and 60% epiboly in DS4.

Developmental timing. Embryos of one stage are collected by morphology: each dish was inspected visually, and 20–40 embryos were taken once they reached the desired stage [7]. Two pools of the same nominal stage can therefore differ by some minutes of development. We use two independent indicators of developmental time, each expressed in units of one "stage step" (the difference between the pair and an adjacent stage of the same batch, so that, e.g., +0.25 means that b is a quarter of a stage step older than a):

- **Ribosomal fraction**: the share of UMIs from the 79 ribosomal protein genes. Across the atlas, it increases monotonically with developmental time, from approx. 2% at high to 21–24% at bud, and within a library, it does not correlate with the depth of a cell (median Spearman correlation −0.05; `results/tables/library_ribosomal.tsv`). However, it no longer increases consistently between 3- and 6-somite.
- **Stage-marker score**: the ratio of UMIs in genes that are more than 2× higher in the older stage to UMIs in genes that are more than 2× higher in the younger stage (in both libraries of the pair; ribosomal genes excluded). For high and oblong, the reference stage is dome, so this score measures the progress of zygotic genome activation.

We consider timing supported only if both indicators agree (Table 5). This is the case for **bud DS2, where library a is older** (by 0.5 and 0.2 stage steps). This is consistent with library a containing more cells expressing the muscle genes of the adaxial cells (*myod1*, *myl10*, *actc1a*), which start to express muscle genes during late gastrulation, and fewer cells expressing *sox19b*, an early ectodermal SoxB1 gene. The low detection rates of these genes (at most 21%) suggest that this is a difference in the number of such cells, as expected if library a is slightly more advanced. Removing the 1,000 most different genes does not eliminate the difference (AUC 0.78), so timing explains at most part of it. In 50% epiboly DS4 (library b older) and oblong (library a older), the indicators agree in direction, but the marker score shift is small. At high and 30% epiboly, they disagree. In particular, at high, the zygotic score suggests that library b is *younger*, while its ribosomal fraction is higher. At the earliest stages, development changes the ribosomal fraction by less than one percentage point, and in both the high and oblong pairs, the library with the higher ribosomal fraction is the much deeper one, so we suspect a technical component and do not trust this indicator before dome. For 6-somite, neither indicator is usable, because neither changes consistently between 3- and 6-somite.

Unexplained differences. For 30% epiboly and 6-somite, we found no specific explanation: both are separable (AUC 0.97 and 0.89), but none of the three explanations is clearly supported. Furthermore, this analysis cannot separate a difference in cell type composition from a shift in expression within cell types. For example, the higher expression of EVL keratins (*krt8*, *krt18*, *cldne*) in oblong b comes with higher detection rates (4–7% vs. 12–21%), so library b probably simply contains more EVL-like cells. We will revisit this distinction once the cell types are identified.

6 Method: aligning libraries

Section 5 showed that libraries cannot be pooled: even two libraries of the same stage and batch are separable. Still, we need to compare cells across libraries, e.g., to ask whether a population found in one library is also present in another. We therefore *align* the libraries: we remove what differs between them on average, and keep how their cells differ from each other. This section defines the alignment, tests whether it does what we need, and states when it may be used.

Table 5: Summary of the a/b differences. On/off genes are detected in at least 40% of the cells of one library and at most 10% of the other (after depth matching). Ribosomal % is the median share of UMIs from ribosomal protein genes. Shifts are in stage steps, positive if library b is older; n/a if the indicator does not change between the pair and the reference stage. Timing is "a/b older" if both shifts agree and are at least 0.1, "weak" if they agree but one is smaller, and "conflicting" if they disagree. Reference stages: dome (high, oblong), shield (30% epiboly), 60% epiboly DS4 (50% epiboly), 90% epiboly DS2 (bud) and 3-somite (6-somite).

Pair	Cells (a/b)	On/off	Ribo. % (a/b)	Ribo. shift	Marker shift	Timing
high DS5	263/48	2	1.8/2.7	+0.58	−0.14	conflicting
oblong DS5	72/128	2	2.9/2.0	−0.70	−0.04	weak, a older
30% epiboly DS5	448/1,058	0	6.6/6.5	−0.10	+0.07	conflicting
50% epiboly DS4	1,567/1,580	3	7.8/9.1	+0.33	+0.10	weak, b older
bud DS2	250/506	0	24.3/21.3	−0.49	−0.20	a older
6-somite DS5	2,375/2,085	0	29.4/25.9	n/a	n/a	n/a

Alignment. Every library is transformed separately. A UMI count y of a gene in a cell is replaced with its Pearson residual:

$$r = \frac{y - s_i m}{\sqrt{s_i m + \varphi_t(m)\,(s_i m)^2}},$$

i.e., the deviation of the count from what the negative binomial model of the library (Section 4) expects for a homogeneous gene, in units of the expected standard deviation. Here m is the mean of the gene in *this* library, s_i is the sequencing depth of cell i relative to the library average, and $\varphi_t(m)$ is the typical dispersion of a gene of mean m in the library. This is the transformation of sctransform [15], with our own dispersion trend. Residuals are clipped at ± 10, the same limit for every library (a limit that depends on the library size makes two unequal halves of one library look different).

The residuals do most of the alignment. The model is fitted to each library separately, so the expected count $s_i m$ is built from the library's own gene mean. A gene that is, on average, twice as abundant in library A as in library B has twice the m in A, and its residuals measure each cell's deviation from its own library's mean. Since the s_i average to 1, the numerators $y - s_i m$ sum to exactly zero over the library's cells, so every gene has a mean residual close to zero in every library before anything else is done. The difference between library averages, whatever its cause (batch, a/b, timing, but also the composition of the library), is thereby removed. What remains are deviations from the average, scaled by the noise expected for the gene's mean, so they are comparable between libraries of different depth and dispersion. Fitting one model to the pooled libraries would instead keep the differences between library averages in the residuals.

Centring. The mean residual is close to zero, but not exactly zero: the denominator gives cells of different depth different weights, and the clip cuts off the largest values of genes expressed in only a few cells. We therefore subtract every gene's mean residual within its library (*centring*). The correction is small: before centring, the mean residual of a heterogeneous gene is a median of 0.02 of its standard deviation (90% of the genes below 0.11; checked at 50% epiboly and bud), and only a few sharply restricted genes reach 0.6–0.85. After centring, the average of every gene is exactly zero in every library. Centring shifts all cells of a library by the same amount, so it does not change the distances between them; the residual transformation does change them, as does any normalization.

By *alignment* we mean both steps together: per-library residuals, then centring. We use the genes heterogeneous at the stage; one principal component analysis (30 components) of all libraries of the stage gives shared axes.

What we test. We compare the aligned data with the unaligned embedding of Section 5 (joint log-normalization and PCA of the compared libraries, on the same genes), in four ways (Figure 8):

1. **Libraries of a stage mix.** For every pair of libraries of a stage, the share of cells whose 20 nearest neighbours are significantly unbalanced between the two libraries (as in Section 5), compared with a random split of one library. Aligned, the random splits give 1–6% (as unaligned), the a/b pairs 1–28% (median 18%; unaligned 12–82%, median 39%), and pairs from different batches 1–36% (median 18%; unaligned 4–95%, median 63%). The alignment removes most, but not all, of the difference: in about half of the pairs, 20–36% of the neighbourhoods remain unbalanced (e.g., 50% epiboly DS4 a/b, 6-somite a/b, 50% epiboly DS2 vs. other batches).

2. **Stages stay apart.** Pairs of libraries of different stages from the same batch (so that the batch does not contribute), 800 cells each. Aligned, stages approx. 1 h apart are blurred (11–44% unbalanced, median 29%; unaligned 35–97%), stages 1.3–2 h apart mostly separate (57–86%, median 73%), and stages more than 2 h apart separate nearly as well as without alignment (78–96%, median 92%). The alignment removes the stage averages too, but cells of distant stages still have no neighbours in common: their populations and gene combinations differ.

3. **Structure within a library is kept.** For every cell, the share of its 20 nearest neighbours in the shared PCA that are also among its 20 nearest neighbours in a PCA of its own library alone: 58–100% (median 79%). The rest is due to the shared choice of axes. The unaligned joint embedding keeps much less (32–73%, median 47%), because its axes also describe the differences between libraries.

4. **Cell types find each other.** Six populations with unambiguous markers, at the stages where they are unambiguous (EVL up to shield, primordial germ cells, endoderm, notochord from 75% epiboly, hatching gland from bud, and the p53-like cells), called in every library from a marker score (positive if above the 99.9% quantile of the scores of random gene sets with the same expression levels); the marker genes are removed from the features. For every pair of libraries, each cell of one library takes its 20 nearest cells of the other, and we ask how well the share of marker-positive neighbours predicts the cell's own call (AUROC). Aligned, the median AUROC is 0.96 (EVL), 0.99 (endoderm, notochord), 0.98 (hatching gland), 0.88 (p53-like) and 0.71 (PGCs, only approx. 10 cells per library), the same as unaligned. Populations that are distinct in every library are found by both; the alignment does not lose them.

Pictures. Figure S5 shows the aligned libraries of every stage; the libraries overlap. Figure S6 shows, on the same coordinates, the marker-called populations of the cell type check in every library (the marker genes are not among the features, so the islands are not drawn by the markers themselves). Where a population is distinct, it forms the same island in every library of the stage: the EVL from dome to 50% epiboly, the p53-like cells at 50% epiboly, the endoderm from 50% epiboly to bud, the PGCs at 75% epiboly, the notochord from 75% epiboly and the hatching gland from bud. Elsewhere, the called cells are partly scattered: part of the EVL at 30% epiboly and shield, the p53-like cells after 60% epiboly, and the PGCs at most stages (approx. 10 cells per library), which matches their lower transfer AUROC (Figure 8d). Figure 9

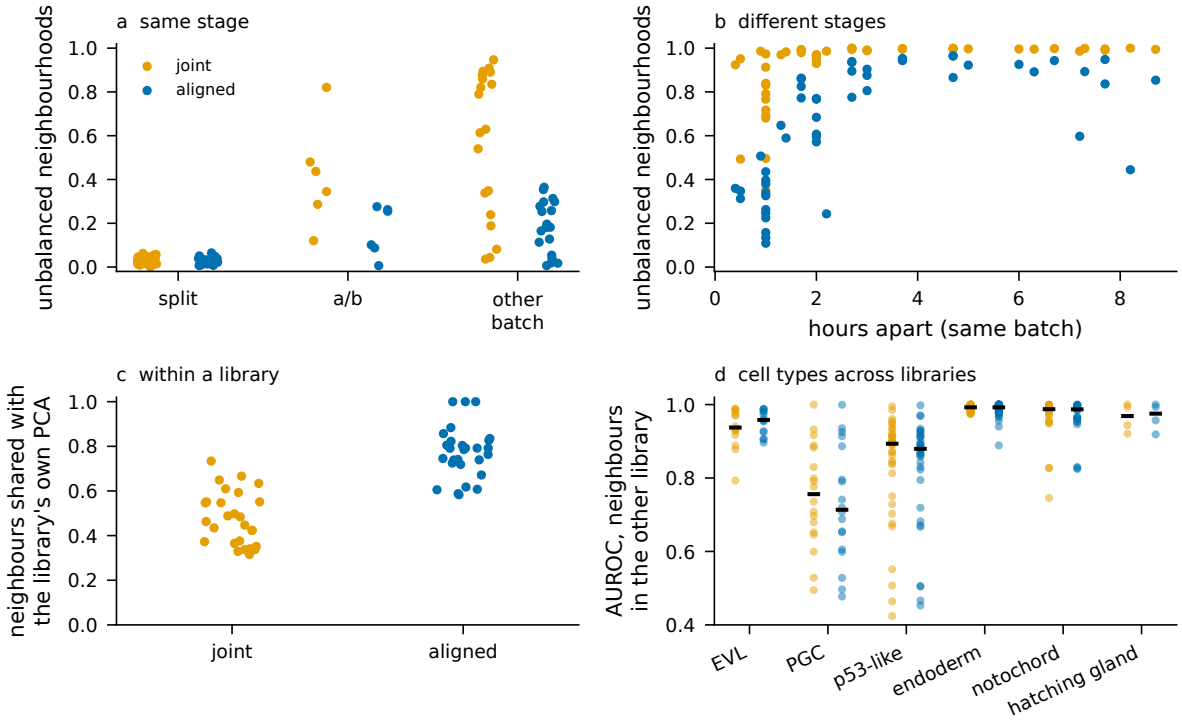


Figure 8: Checks of the alignment (blue) vs. the unaligned joint embedding (orange). (a) Share of cells with unbalanced neighbourhoods for pairs of libraries of one stage, and for random splits of one library. (b) The same for libraries of different stages of one batch, by the time between the stages. (c) Share of a cell's 20 nearest neighbours that are also its neighbours in a PCA of its own library. (d) AUROC of the share of marker-positive neighbours in another library of the stage, per population (black: median).

shows the whole atlas, aligned and unaligned. Unaligned, the atlas is ordered by stage (the DS5 libraries, which hold the earliest and the latest stages, sit at both ends). Aligned, the batches overlap (panel b), and the stages form a gradient rather than separate regions: centring removes the stage averages, so stages separate only where their cells differ, and adjacent stages partially overlap. The aligned atlas is therefore not a developmental time line, and its picture should not be read as one. The other side of the same effect is shown by the marker populations (panels c and f): aligned, each population forms one island across all the stages where it is called (e.g., the endoderm from 50% epiboly to bud, the notochord from 75% epiboly on, and a compact island of p53-like cells), while unaligned, the endoderm is split between early and late stages and the p53-like cells are scattered over the stage gradient. The aligned atlas groups cells by what they are rather than by when they were sampled.

When to use which.

- **Aligned data** for everything that compares *cells* across libraries: finding a population of one library in another, joint neighbourhoods and pictures of a stage, and links between populations of nearby stages. Even here, two caveats: 20–36% of the neighbourhoods of some pairs remain unbalanced, so conclusions should not require libraries to be interchangeable, and stages approx. 1 h apart are partly blurred, so a match between nearby stages is weaker evidence than a match within a stage.
- **Original libraries** for everything about library *averages*: library-level programs, pseudobulk expression, a/b differences, timing, and the composition of a library (how many

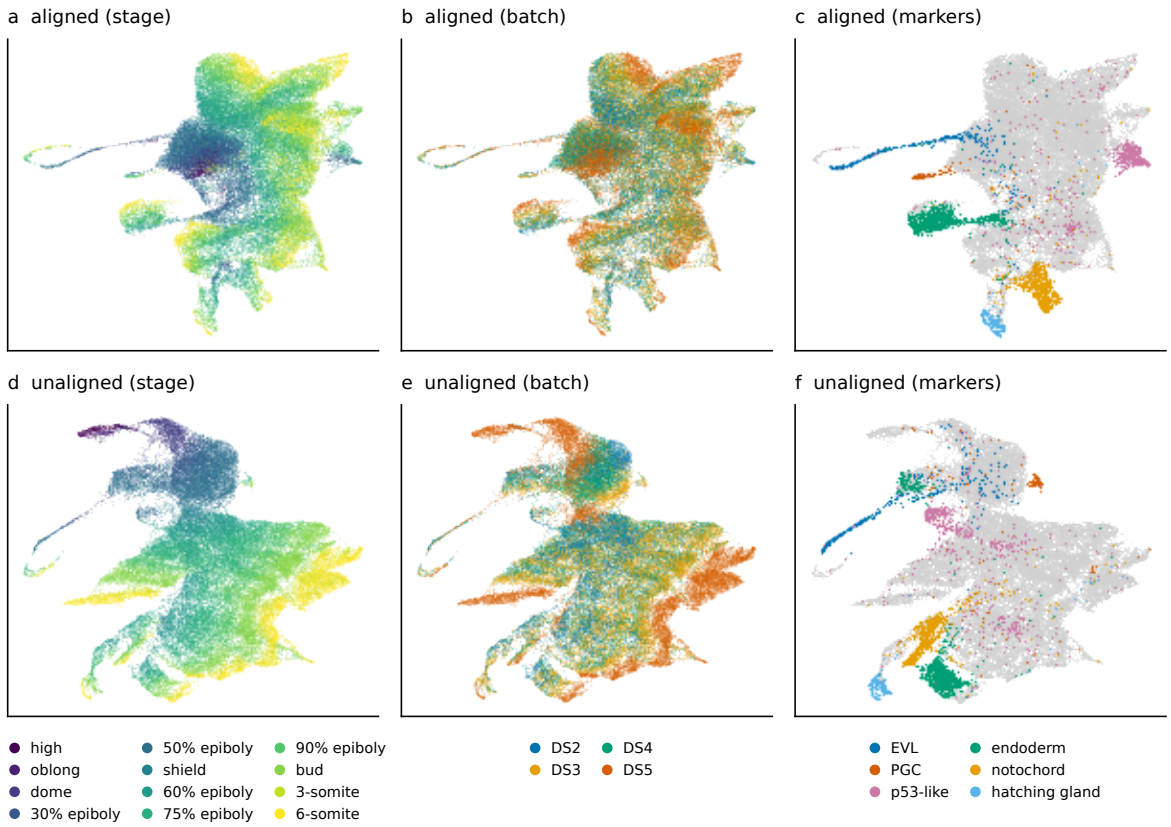


Figure 9: UMAPs of the whole atlas on the genes heterogeneous at any stage (marker genes excluded). Top (a–c): aligned (per-library residuals, centred); bottom (d–f): unaligned (log-normalized, scaled). Colour: stage (a, d), sequencing batch (b, e), and the marker-called populations of Figure S6 (c, f; grey: no call, or a stage where the population's markers are not used).

cells a population has). The alignment removes all of these by construction.

7 Result: expression programs describe the cell-to-cell variation

An *expression program* is a set of genes associated with one cell type, tissue, cell state or signalling pathway at one stage, e.g., the genes recorded in the notochord at bud, or the feedback targets of BMP. Programs may share genes, and a cell may have several active programs. A program is a hypothesis about co-expression, not a cell type: whether its genes vary together in the cells is tested below. In this section, we define several sets of programs, from the data and from prior knowledge, and ask (i) whether each program is visible in the data, and (ii) how much of the variation between cells each set explains.

7.1 Program sets

All sets except the final curated one are stage-specific. A program of a stage contains only genes measured at that stage (at least 10 UMIs), and is capped at 30 genes (the number of genes listed per module by [7]). The sets (Table 6; `scripts/16_program_sets.py`):

- **Farrell et al. modules**: the gene modules that [7] found by non-negative matrix factorization in each stage of this dataset (their Table S2: the 30 listed genes of each module, with weights). They were found in these cells, so they have an in-sample advantage.

- **ZFIN terms**: one program per anatomy term of ZFIN [1]: the genes with an in situ record in the term whose stage interval covers the atlas stage (as in Section 4). Broad terms (whole organism, unspecified, ...) and the yolk are left out, a term needs at least 3 genes, and a term whose genes are mostly (≥80%) those of a larger term is merged into it. Two versions: records published before 2018, and all records.

- **ZFIN terms, heterogeneous**: the same, restricted to the genes heterogeneous at the stage (Section 4).

- **Signalling**: readouts of seven pathways (BMP, Nodal, Wnt, FGF, retinoic acid, Notch, Hedgehog), i.e., feedback inhibitors and direct targets rather than ligands or receptors, whose expression measures pathway activity, not the ability to respond (`data/curated/signalling_programs.tsv`; e.g., *szl*, *bambia*, *id1* for BMP, *lft1*/*lft2* for Nodal, *sp5l*, *axin2* for Wnt, *spry4*, *dusp6*, *etv4*/*etv5b* for FGF, *cyp26a1* for RA, *her4*, *nrarp* for Notch, *ptch2*, *gli1* for Hedgehog). Restricted to heterogeneous genes, at least two per pathway. Many readouts are also identity markers (*id1* and the *vent* genes of the ventral ectoderm, *lft1* of the margin), so a signalling program partly overlaps the programs of the tissues that receive the signal.

- **Cell states**: maternal transcripts, the pluripotent blastomere state, S and G2/M phases, heat shock and the p53 response (our earlier literature table, `data/curated/archive/`). These are not anatomy and are absent from ZFIN's anatomy terms.

- **Curated, stage-specific genes** (first draft): the ZFIN terms (all years) grouped into programs that keep their name across stages (`data/curated/zfin_anatomy_programs.tsv`, e.g., shield, embryonic shield and organizer into *dorsal organizer*); a gene belongs to a group if it has a record in the group at the stage and the group holds at least a third of its records; restricted to the genes heterogeneous at the stage, and capped at the 30 genes with the most records at the stage; plus the signalling and state programs. ZFIN's maternal and blastomere terms are left out: at later stages, they collect ventral and marginal genes (*foxi1*, *bmp2b*, *gata2a*); the literature state programs describe these states instead.

- **Curated** (fixed genes; the set used in the rest of the report): the same programs, but with one gene list for all stages. In the stage-specific version, only the name persists: the gene list of a program is chosen anew at every stage (heterogeneity at the stage, the cap, ZFIN's coverage of the stage), so that, e.g., the EVL program has 69 different genes over its 10 stages, none of them at all stages, and the median anatomy program shares fewer than half of its genes between stages. A change in its score from one stage to the next would partly reflect a change in its genes, which is not acceptable for a variable of a model. Here, every program has a *window* of stages, from the first to the last stage at which its stage-specific version exists, and one gene list for the whole window: the group's records are pooled over the window, a gene must be heterogeneous at some stage of the window, and the genes are ranked by their mean heterogeneity tier over the window (0 = not heterogeneous, 1–3 = weak to strong; Section 4), so that genes that vary throughout the window come first, and then by their number of records; the first 30 are kept. Literature evidence thus decides whether a gene belongs to a program, the data only which of the members are kept. Groups whose identity changes over the window are split (column `fixed_program` of the mapping table): marginal mesendoderm into *margin* (germ ring, margin) and *mesoderm* (mesoderm, hypoblast), the notochord into *axial mesoderm* and *notochord*, the prechordal plate into *prechordal plate* and *hatching gland* (polster, hatching gland), and the neural ectoderm into *presumptive neural* and *neural plate*. Signalling programs keep the pathway's readouts that are heterogeneous at some stage of the window (the stages with at least two heterogeneous readouts); the state programs are the same at every stage anyway. At a

Set	Programs per stage	Genes per program	Genes per stage	Genes
Farrell et al. modules	10–45	30	249–514	887
ZFIN terms, pre-2018	1–84	17	7–718	1,148
ZFIN terms, all	1–84	18	7–737	1,195
ZFIN terms, pre-2018, het.	2–78	12	6–473	691
ZFIN terms, all, het.	2–78	11	6–481	706
Signalling	1–6	3	2–24	30
Cell states	6–6	10	40–51	51
Curated, stage-specific genes	7–33	8	49–453	688
Curated, fixed genes	8–36	10	55–522	562

Table 6: Program sets: programs and distinct genes per stage (range over stages), median genes per program, and distinct genes over all stages; het.: restricted to heterogeneous genes.

given stage, a median of 71% of a program's genes are heterogeneous (49% at the worst stage of the window).

The pre-2018 and all-years ZFIN sets are nearly identical (the early expression records are mostly older), so we report the all-years version. ZFIN has few localized terms before 50% epiboly (1–7 per stage), because most early records are of the whole embryo.

Scores. The score of a program in a cell is the mean of the Pearson residuals of its genes under the model of the cell's library (Section 6), centred per library (weighted for the Farrell et al. modules). The residuals already account for sequencing depth, so, unlike the usual gene-set score [47], no control genes are needed, and the score has the same meaning in aligned and per-library data. Since the residuals are centred per library, the score describes differences between the cells of a library; a program that is active in a whole library is a constant within it.

7.2 Is a program visible in the data?

If the genes of a program vary together in the cells, the program is visible in the data. We measure this by the coherence

$$R = \frac{n \operatorname{Var}(\text{score})}{\text{mean variance of a gene's residuals}} = 1 + (n-1)\,\bar{\rho},$$

where n is the number of genes and $\bar{\rho}$ their mean correlation; $R = 1$ if the genes vary independently. The obvious null, shuffling every gene independently across the cells, is too weak: random sets of heterogeneous genes pass it in 74% (50% epiboly) and 27% (bud) of cases, because heterogeneous genes vary along a few shared axes of the stage (e.g., margin vs. animal pole). We therefore require a program to be more coherent than 95% of 500 random sets of heterogeneous genes of the same size, in the same library, and call it **supported** at a stage if this holds in at least half of the stage's libraries (`scripts/17_program_tests.py`).

The 95% quantile of the random sets is itself an estimate: with 50 random sets (our first version), approx. 6% of the calls changed with the random seed, most of them small programs close to the threshold, so we use 500.

From 60% epiboly on, 48–81% of the curated programs are supported (with stage-specific genes: 68–82%; Farrell et al.: 53–81%, ZFIN terms: 55–73%, restricted to heterogeneous genes: 63–80%). Fixed gene lists cost coherence: at a given stage, part of a program's genes are not heterogeneous there and only dilute the score; the difference is largest at 75% and 90% epiboly (48% vs. 68% supported). The unsupported ones are mostly expected: maternal transcripts (gone), heat shock (uniform), the p53 response at most stages from 75% epiboly, the RA readouts, and broad programs that cover several populations: *mesoderm*, *neural plate* and *presumptive neural*

at nearly every stage, *ventral BMP*, and the dorsal forerunner program. Up to shield, only 12–27% of the curated programs are supported (Farrell et al. modules: 18–40% at dome to shield). Here, the test has little power rather than finding spurious programs: before gastrulation, most heterogeneous genes vary along the same few axes, so random sets are nearly as coherent as real programs, and removing the unsupported programs loses part of their predictive power (below).

Library-level programs. The maternal and blastomere programs act on whole libraries: their pseudobulk expression (median over genes, relative to the highest library) falls from 0.87 (high) to 0.23 (dome) and below 0.1 from 30% epiboly (maternal), and is 0.70–0.78 from 30% epiboly to shield and below 0.2 from 90% epiboly (blastomere). Within a library, they are not coherent up to shield; the blastomere program becomes supported from 60% epiboly, when some cells (neural) keep it and others have lost it. Such programs must be evaluated on the original libraries (Section 6).

7.3 How much of the variation does each set explain?

We repeat the prediction test of our earlier analysis on all cells of each stage: evaluation genes, never part of any program of any set at the stage, are predicted from a cell's program scores, and the test log-loss is compared with a floor (sequencing depth and library) and a ceiling (the best model found, including the detection frequency among a cell's nearest training cells). The explained fraction is $(\text{loss}_{\text{floor}} - \text{loss}_{\text{set}})/(\text{loss}_{\text{floor}} - \text{loss}_{\text{ceiling}})$. Scores are smoothed over each cell and its 10 nearest neighbours in its library, and the target is either a gene's detection in the cell or its detection frequency in the cell's neighbourhood. For every program, a random set of genes with the same expression levels gives the random reference. Two groups of evaluation genes: 400 random measured genes (*all*), and up to 400 heterogeneous genes.

Little of the variation of random genes can be explained by anything: the best model removes at most 7% of the floor loss (neighbourhood target), and we report only stages where at least 4% can be explained. Heterogeneous genes are much more predictable (26–36% of the floor loss from dome on, except shield: 7%; 6% at high and oblong). The ceiling is the best of all fitted models, which is usually one of the feature sets (the Farrell et al. modules up to shield, the smoothed principal components or all features later), rarely the nearest-neighbour model; an explained fraction of 1 thus means "as good as the best model we tried". The main result (Figure 10, neighbourhood target, heterogeneous genes, dome to 6-somite):

- **Every set explains far more than random genes.** Explained fraction: Farrell et al. 0.98–1.00, ZFIN terms 0.87–0.99, curated 0.91–0.99, vs. 0.30–0.83 for random sets matched to the curated programs. Relative to random ((set − random)/(1 − random)): curated 0.67–0.92, ZFIN terms 0.48–0.95, Farrell et al. 0.83–1.00. The same holds for single-cell detection, and for the random genes where defined (curated 0.82–0.96 vs. random 0.31–0.66).
- **Restricting to heterogeneous genes costs nothing.** ZFIN terms restricted to heterogeneous genes explain the same (0.89–0.99) with a third fewer genes.
- **The curated draft is as good as the ZFIN terms with fewer programs.** With 10–36 programs per stage from dome on (ZFIN terms: up to 84), it explains 0.95–0.99 before shield (ZFIN terms: 0.87–0.99, where ZFIN has few terms) and 0.91–0.96 from shield on (0.90–0.96). The Farrell et al. modules remain the best, as expected from their in-sample advantage.
- **Fixed gene lists cost no predictive power.** The curated programs with one gene list per window explain as much as with stage-specific genes (0.91–0.99 vs. 0.89–0.99), and as much relative to random sets (median 0.85 vs. 0.84; more at shield and 75% epiboly, less at 3- and 6-somite).

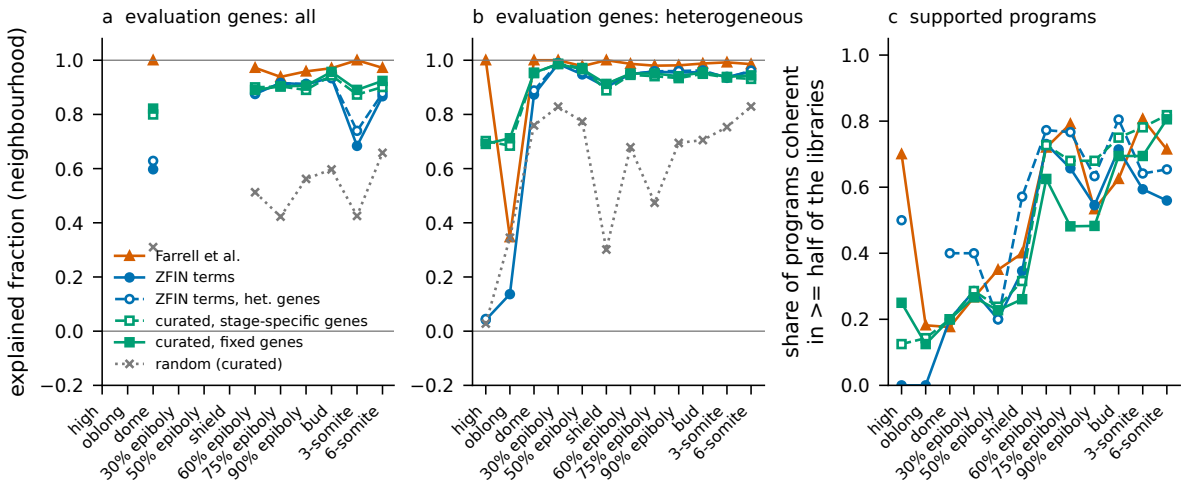


Figure 10: Program sets. (a, b) Explained fraction of the neighbourhood detection of evaluation genes (a: random measured genes, only where at least 4% of the floor loss can be explained; b: heterogeneous genes), 0 = depth and library only, 1 = best model. Green: curated programs with fixed (solid) and stage-specific (dashed) gene lists; grey: random gene sets matched to the curated programs. (c) Share of programs that are supported (more coherent than random heterogeneous genes in at least half of the stage's libraries).

- **Unsupported programs matter early, not late.** From 60% epiboly on, the supported curated programs alone (13–29 per stage) explain 0.86–0.94 (all curated: 0.94–0.96). Up to 50% epiboly, with only 2–5 supported programs, they explain less (dome: 0.84 vs. 0.95; 50% epiboly: 0.92 vs. 0.97), confirming that the coherence test is too strict there.
- **Signalling and states alone** explain 0.21–0.68 and 0.32–0.87: they are a few programs each, which describe a part of the variation that overlaps the identity programs.

7.4 Which programs co-occur with which signals?

The signalling programs measure where a pathway is active at a stage, so their correlation with the other programs tells us which populations receive which signal, and when (Figure 11; `scripts/21_signal_coupling.py`). For every stage, we correlate the smoothed program scores over the cells (within libraries, since scores are centred per library). Many readouts are also identity markers (e.g., *lft1* and *ndr1* in the margin, *id1* and *bambia* in the organizer and prechordal plate programs), so we first remove the readout genes from the other programs (16 genes, 147 program–stage memberships); otherwise, shared genes would correlate programs by construction. A correlation says that a pathway's readouts are expressed in the cells of a program at that stage, not that the signal causes the program; smoothing over neighbouring cells inflates the absolute values, so the changes over time and the signs are more informative.

Most couplings agree with what is known about these pathways in the zebrafish gastrula:

- **Nodal** is coupled to the margin (0.89–0.93 from dome to 50% epiboly) and the endoderm (0.89–0.93 at 30% and 50% epiboly), and the coupling fades after shield (margin 0.65 at shield, 0.35 at 75% epiboly, approx. 0 from bud): Nodal signals are required for endoderm and most mesoderm [8, 13], and the duration of signalling during the blastula and early gastrula decides the fate [16]. The coupling then moves along the axis: dorsal organizer (0.74–0.87 from 30% to 60% epiboly), prechordal plate (0.58–0.69 from 50% to 75% epiboly), axial mesoderm (0.62–0.69 at shield and 60% epiboly) and floor plate (0.67 at 3-somite), as expected from the highest Nodal requirement of the prechordal plate [12]

and the expression of Nodal ligands and *lefty* in axial tissues [35, 42]. The blastomere program is anti-correlated (-0.55 to -0.68 from dome to 60% epiboly): cells that receive Nodal have lost the pluripotency genes.

- **BMP** is coupled to ventral and non-neural fates (non-neural ectoderm 0.52–0.58 at 75% epiboly to bud, ventral BMP program up to 0.58, placodes 0.49–0.52, tailbud 0.64–0.72 from bud) and anti-coupled to the neural plate (-0.37 to -0.57 from 50% epiboly to bud), the presumptive neural program, the forebrain and the hindbrain: the classic BMP gradient of dorsoventral patterning [17, 21, 30], with the tail requiring BMP [32]. The anti-coupling with the neural plate weakens after bud (-0.19 at 6-somite), as BMP patterning proceeds along the body axis [46].

- **Wnt** is coupled to the margin and the posterior mesoderm (margin 0.72–0.82 from 30% to 60% epiboly, presomitic mesoderm 0.73–0.74 at 50% epiboly and shield, mesoderm 0.60–0.62, tailbud 0.40–0.73 from bud) and anti-coupled to the forebrain and eye (-0.39 to -0.50) and the non-neural ectoderm: Wnt8 at the ventrolateral margin is required for ventrolateral mesoderm and posterior fates [25, 43], and the forebrain requires Wnt to be off [20, 18]. The coupling with the endoderm (0.61–0.67 early) more likely reflects their common position at the margin.

- **FGF** (measurable from 60% epiboly to bud only) is coupled to the margin (0.50–0.64), tailbud (0.65), and presomitic mesoderm (0.37–0.43), and anti-coupled to the forebrain, hindbrain and non-neural ectoderm: FGF at the margin is required for trunk and tail mesoderm [11, 5], where its feedback genes are expressed [9, 33].

- **Notch** is coupled to the tailbud (0.40–0.49 from bud) and the posterior margin (0.42 at bud), but not clearly to the presomitic mesoderm ($r < 0.3$), although the segmentation clock depends on Notch [19]; the clock genes oscillate, so at any moment only part of the presomitic cells express the readouts.

Some expected couplings are missing, and some readouts are not interpretable:

- Notch is not coupled to the primary neurons, although *her4* is a Notch target in the proneural clusters [40]; FGF is not coupled to the midbrain–hindbrain boundary [36]. Both pathways have few, lowly detected readouts at these stages.

- The retinoic acid readouts measure something else: *cyp26a1*, an RA-degrading enzyme, is expressed where RA must be kept low (tailbud, anterior neural plate) and is regulated by FGF and Wnt as well [23, 29]. The coupling with the tailbud (0.64–0.66 at 3- and 6-somite) is therefore not RA activity, and the expected coupling with the hindbrain is not seen.

- Hedgehog readouts (*ptch1/2*, *gli1*, *hhip*) are never heterogeneous, so there is no Hedgehog program, and its couplings with the floor plate and adaxial cells cannot be tested.

- Unexplained: Nodal with maternal transcripts (0.45–0.59 from 30% epiboly to shield) and with heat shock (0.46 at 60% epiboly), BMP with maternal transcripts (-0.44 to -0.58 from high to dome, when the BMP readouts are barely expressed), and Wnt and FGF with the presumptive neural program (0.56 at shield and 0.54 at 60% epiboly), which may reflect the posterior part of the neural ectoderm, next to the margin. The first two may be technical.

7.5 How many activity levels does a program need?

A Boolean or multi-valued model describes every program in a cell by one of a few levels. We ask how much of what the continuous scores say about a cell survives when every curated program is reduced to k levels (`scripts/22_program_levels.py`). We reuse the prediction test above with the same evaluation genes, cells, split, smoothing, floor and ceiling, so the numbers are on the scale of Figure 10; the only change is the features:

- **k levels** (k = 2–6): the smoothed scores of each program are cut into k levels by 1-D k-means of the training cells, pooled over the stage's (aligned) libraries, i.e., one set of thresholds per program and stage, as a model would use; every cell gets the level of its nearest centre.
- **Boolean (null)**: active if the score is above the 99.9% quantile of the gene-wise permutation null of the library (the "active" call of Section 8), and **3 states**: additionally depleted below the 0.1% quantile.

A program with k levels enters the regression as $k - 1$ indicators "level $\geq v$". This allows any effect of every level, but the regularization of the regression shrinks a level towards the one below it rather than towards the lowest; indicators "level $= v$" span the same models but are worse with many levels and few cells (e.g., 6 levels at shield: 0.39 vs. 0.61 of the continuous result). The levels of a program also allow effects that are not linear in its score, so a discretised set can explain slightly *more* than the continuous scores.

Result (Figure 12; neighbourhood target, heterogeneous evaluation genes, dome to 6-somite; "kept" = explained fraction relative to the continuous scores, which explain 0.91–0.99):

- **Two levels keep about 90%** (median 0.93, range 0.77–0.98). **Three levels keep nearly everything** (median 0.98, 0.81–0.99), and four or more are as good (median 0.97–0.98); from 60% epiboly on, at the stages with several libraries, 3 levels keep 0.97–0.99 and 4 levels 0.98–1.01.
- **Where the threshold lies matters as much as the number of levels.** The Boolean call from the permutation null keeps only 0.86 (0.72–0.91), less than two k-means levels: its strict threshold marks only cells that are clearly active, and cells with an intermediate score fall into "off". Adding the depleted state recovers part of it (3 states: 0.93, 0.62–0.96), but less than three k-means levels.
- **Shield loses most.** At shield (one library, 1,046 cells), 2 levels keep 0.77, 3 levels 0.81, and more levels do not help (4 levels: 0.66): the thresholds and the effects of the levels are fitted on few cells, and the test itself is weak there (the best model removes only 7% of the floor loss). The other single-library stages lose little (3 levels: dome 0.97, 3-somite 0.96).
- **The supported programs alone, with 3 levels**, keep 0.90–0.98 of the continuous curated set from 60% epiboly on, so restricting a model to them costs up to a tenth; with fixed gene lists, fewer programs are supported (Section 7), and the broad unsupported ones (mesoderm, neural plate) still carry information.
- **Random measured genes** show the same order with larger losses: from 60% epiboly on, 2 levels keep 0.76–0.91, 3 levels 0.80–0.93 and the Boolean call 0.61–0.80, except at 3-somite, where 3 levels keep only 0.66.

Which programs are switches? Separately from the prediction, we measure how much of the variance of a program's smoothed score its k-means levels keep (Figure 12c). Two levels keep a median of 68% of the variance (3 levels: 84%, 4 levels: 91%), much less than they keep of the prediction: what matters for the other genes is mostly which cells are at the top, and the variation within the bulk of the cells matters less. The programs that come closest to a switch (2 levels keep 73–86% of the variance, and the upper level holds 1–10% of the cells) are distinct populations: the EVL, the axial mesoderm and notochord, endoderm, dorsal organizer, prechordal plate, hatching gland and non-neural ectoderm. The most graded are the cell cycle (2 levels split the cells about in half), the uniform states (maternal, p53 response, heat shock), the signalling readouts (Wnt, BMP, RA, Notch; 64–67%), and the broad or unsupported anatomy programs (hindbrain, intermediate mesoderm, dorsal forerunner; 54–60%), as expected for gradients and continuous processes.

Caveats. The levels are computed from scores smoothed over each cell's 10 nearest neighbours, so they describe a cell's neighbourhood rather than the cell itself; levels of unsmoothed scores would be noisier, and we have not tested them. The thresholds are fitted per stage, so a program's levels need not mean the same at two stages, and they are fitted on aligned scores, so they say nothing about differences between whole libraries (e.g., the maternal program). Finally, the test measures what the levels say about other genes, not whether they are the right variables of a dynamical model.

Conclusion for the model. Three levels per program (low / intermediate / high), with thresholds that are placed by the data rather than by a strict significance cut-off, keep practically all of what the continuous scores explain; a Boolean description keeps about 90% with data-placed thresholds and about 85% with the significance call. The exception is shield, the only stage sampled by a single small library at the focus of the model, where even 3 levels keep only 0.81. The two-level description is close to complete for programs of distinct populations, and loses most for graded programs (signalling, cell cycle), which are the natural candidates for more than two levels. This agrees with our earlier analysis, which found that Boolean values lose part of the predictive power and 3–4 levels recover nearly all of it.

Conclusion. Programs from prior knowledge describe the cells nearly as well as the modules found in the data, and far better than random gene sets. Restricting them to heterogeneous genes loses nothing, and a first curated set of 10–36 named programs per stage, with one gene list per program for all stages, is as useful as the ZFIN terms. From 60% epiboly on, half to four in five curated programs are clearly visible in the data, and the rest are either expected to be invisible (maternal, uniform states) or too broad (mesoderm, neural plate). Before 60% epiboly, the variation between cells is dominated by a few shared axes, and our test cannot tell individual programs apart. For a discrete model, three data-placed levels per program keep nearly all of the information of the continuous scores, and two levels about 90% (Section 7.5).

8 Result: cell populations

Clustering a single library always yields clusters, whether or not the cells form distinct groups, and our earlier attempts gave clusters that were poorly separated and did not recur across libraries. Here, a cluster counts as a *population* only if it is found independently in another library. We find populations twice: from the heterogeneous genes, and from the program scores of Section 7; comparing the two tells us which populations the programs describe and which programs are missing.

8.1 Replicated populations

Procedure (`scripts/14_replicated_populations.py`). Every library is clustered on its own (Leiden [45] on a 15-nearest-neighbour graph, resolutions 0.5, 1, 2 and 4), in the aligned space of its stage (Section 6). Clusters of two libraries are compared by neighbour voting, as in MetaNeighbor [4]: every cell of library B takes its 10 nearest cells of library A, and the vote of an A cluster is the share of them in it. The AUROC of the vote for a B cluster (its cells vs. the rest of B) says how well the A cluster picks it out. Two clusters *match* if each is the other's best counterpart and each still wins (AUROC ≥ 0.8) when compared with the runner-up only, which separates a true counterpart from a merely related cluster. Populations are the groups of matching clusters; populations of different resolutions that share at least half of their cells are merged, and a population is **robust** if found at two or more resolutions. Stages with a single library (dome, shield, 3-somite) are matched against the libraries of the adjacent stages, which is weaker evidence (Section 6).

Controls. Arbitrary clusters of the same number, built around random cells (so compact, but meaningless), and matched in the same way, give 19–31% as many populations as the real clusters (per resolution, summed over stages; most at resolution 4). With another random seed, 95–97% of the robust populations are found again (Jaccard ≥ 0.5), but only 56% of the populations found at a single resolution.

Result. We find 131 robust populations: none at high and oblong, 3–6 from dome to 50% epiboly, 11–17 from shield to 90% epiboly, 25 at bud, 23 at 3-somite and 17 at 6-somite. Their top genes name known populations without any marker list: e.g., the EVL (dome to shield, and 3- and 6-somite), the p53-like cells, the dorsal organizer, which becomes the prechordal plate (*gsc*, *frzb*) and then the hatching gland (*he1a*, *ctslb*), the notochord (*noto*, *twist2*, *ntd5*), the endoderm (*sox32*, *sox17*), the tailbud and presomitic mesoderm, primordial germ cells (75% epiboly), and, by 3-somite, the eye field, the midbrain–hindbrain boundary, hindbrain rhombomeres, the neural crest, placodes, somites, adaxial and early muscle cells, primary neurons and blood/endothelial progenitors. From shield on, 47–94% of the cells (median library) belong to a robust population; from dome to 50% epiboly, only 13–28% (none at high and oblong). Populations of consecutive stages are linked when each picks out the other in a joint analysis of the two stages (AUROC ≥ 0.9), e.g., the endoderm from 50% epiboly to 6-somite (Figure S7); since the alignment blurs stages approx. 1 h apart, a link means "most similar population of the next stage" rather than proof of descent.

8.2 Populations from programs

Procedure (`scripts/18_program_populations.py`). The same procedure, but every cell is described by the scores of the curated programs that are supported at its stage, smoothed over its 10 nearest neighbours in its library and scaled to unit variance. A program-based population is described by its top programs. A gene-based robust population is *captured* if a robust program-based population shares at least 30% of its cells (Jaccard). If not, it is *named, not separated* if at least two of its top 8 genes belong to one curated program of the stage (the program exists but does not separate the population from its neighbours), and *unnamed* otherwise: a candidate missing program.

Result. The program scores give 1–24 robust populations per stage; the arbitrary clusters give 30–43% as many, more than for the gene-based populations, so program-based populations are less specific. Of the 131 gene-based populations, 69 are captured, 53 are named but not separated, and 9 are unnamed (Figure 13b). With stage-specific gene lists, the numbers were

81, 40 and 10: programs that are broad over their whole window (mesoderm, neural plate) are less often supported, and so separate fewer populations.

- **Named, not separated**: mostly parts of broad curated programs, e.g., the ventrolateral and posterior mesoderm, presomitic mesoderm and tailbud within *margin* and *mesoderm*, the forebrain, hindbrain, midbrain–hindbrain boundary, neural plate border and early neurons within *neural plate*, and subpopulations of the EVL. The curated programs are too coarse to separate these populations; the ZFIN terms (or our population genes) are finer.

- **Unnamed**: the non-neural ectoderm from 60% to 90% epiboly (*dlx3b*, *foxi1*, *tfap2a*, *gata3*, *apoc1*, *apoeb*): ZFIN has no such term at 60% epiboly, and with the records pooled over its window, the fixed non-neural ectoderm program loses *dlx3b* and *foxi1* to the placodes and keeps later epidermal genes (*gata3*, *tp63*, *foxi3a/b*, *grhl1*); two mesoderm populations at 90% epiboly, one with *osr1*, *pax8*, *gata5* and *gata6* (intermediate and lateral plate mesoderm, before the window of these programs) and one with *twist1a*, *cxcl12b* and *ripply1* (paraxial or head mesoderm); a population with *apoc1*, *apoeb* and *crabp2a* at 6-somite; the p53-like cells at shield; and a family of uncharacterized genes (*si:ch211-113a14.**, *zgc:110425*, ...) at dome and shield, which may be technical (Table S5). The EVL at dome and 3-somite, the floor plate/hypothalamus and the posterior *hox* population, which had no program with stage-specific genes, are now named.

Coverage. From 60% epiboly on, 58–95% of the cells (median library) have at least one active program (smoothed score above the 99.9% quantile of a gene-wise permutation within the library), and 47–94% belong to a robust gene-based population (Figure 13a). Up to 50% epiboly, only 1–27% of the cells have an active cell-level program, and 0–28% belong to a robust population: at these stages, most cells differ from each other along graded axes rather than by belonging to distinct populations, and the dominant programs (maternal, blastomere) act on whole libraries.

8.3 Conclusions

- **Alignment.** Pearson residuals under each library's own model, centred per library, make cells of different libraries comparable without making the libraries interchangeable; it is to be used for comparing cells, not library averages.

- **Programs.** All program sets explain far more of the variation between cells than random genes; restricting them to heterogeneous genes costs nothing; and a first curated set of 10–36 named programs per stage, with one gene list per program for all stages, is as good as the ZFIN terms. From 60% epiboly on, half to four in five curated programs are clearly visible in the data.

- **Populations.** Requiring a cluster to recur in another library gives 131 robust, reproducible populations, which name themselves through their genes. The curated programs capture 69 of them as program-based populations. Most of the rest lie inside programs that are too broad (margin, mesoderm, neural plate), and nine have no program.

- **Before 60% epiboly**, cells mostly differ along shared graded axes; programs and populations describe only a minority of the cells, and the main programs act on whole libraries.

- **Next steps for the curated set**: split the remaining broad programs (mesoderm into paraxial, lateral and intermediate parts before bud; neural plate into regions), using ZFIN terms or the gene-based populations; give the early non-neural ectoderm its own genes (*dlx3b*, *foxi1*, *tfap2a*), which pooling over the window assigns to the placodes; add programs

for the remaining unnamed populations; and use about three activity levels per program (Section 7.5).

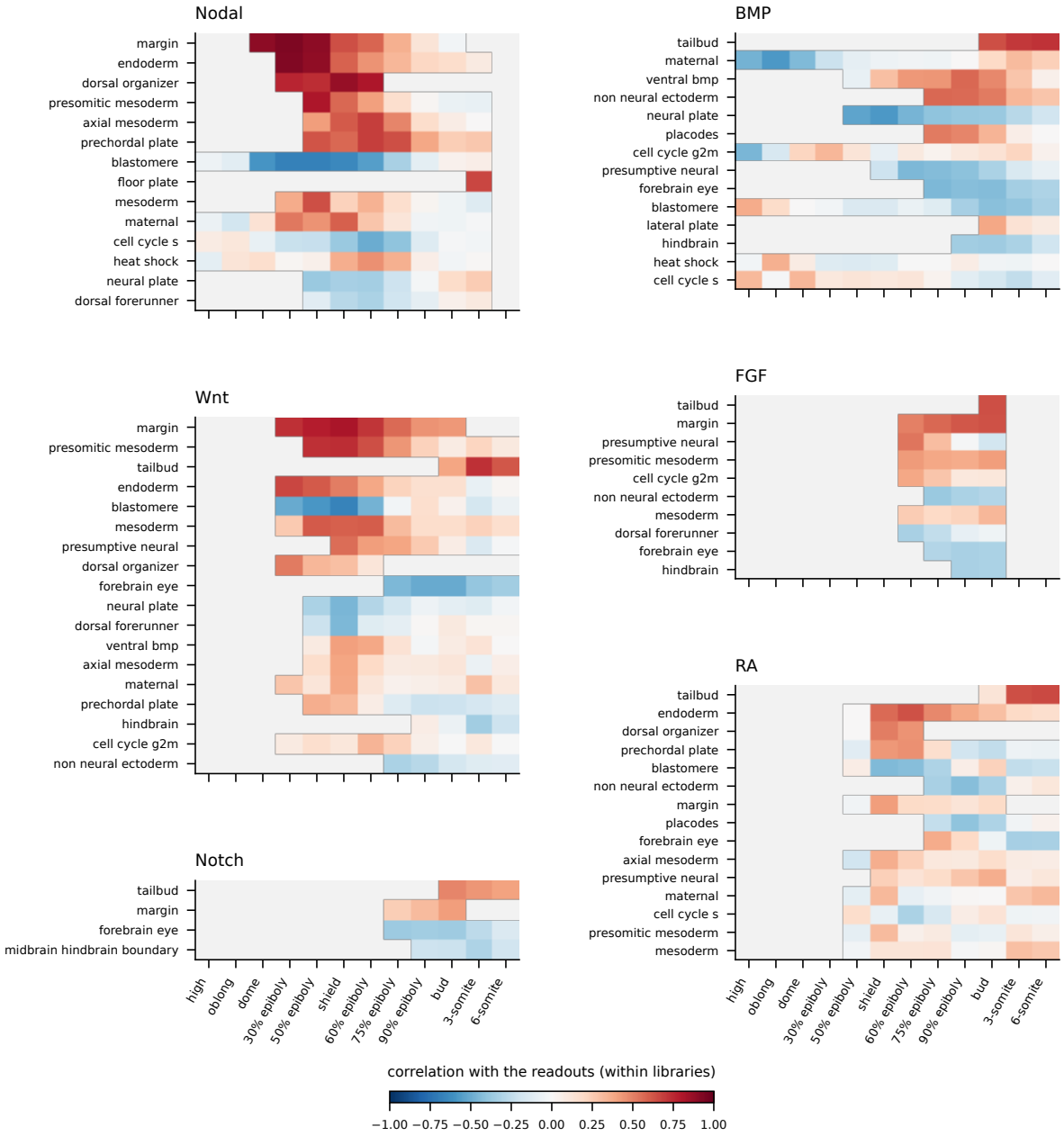


Figure 11: Correlation of the signalling readouts with the other curated programs (smoothed scores, within libraries; readout genes removed from the other programs), per stage. Shown: programs with $|r| \geq 0.3$ at some stage; grey: program or readout not defined.

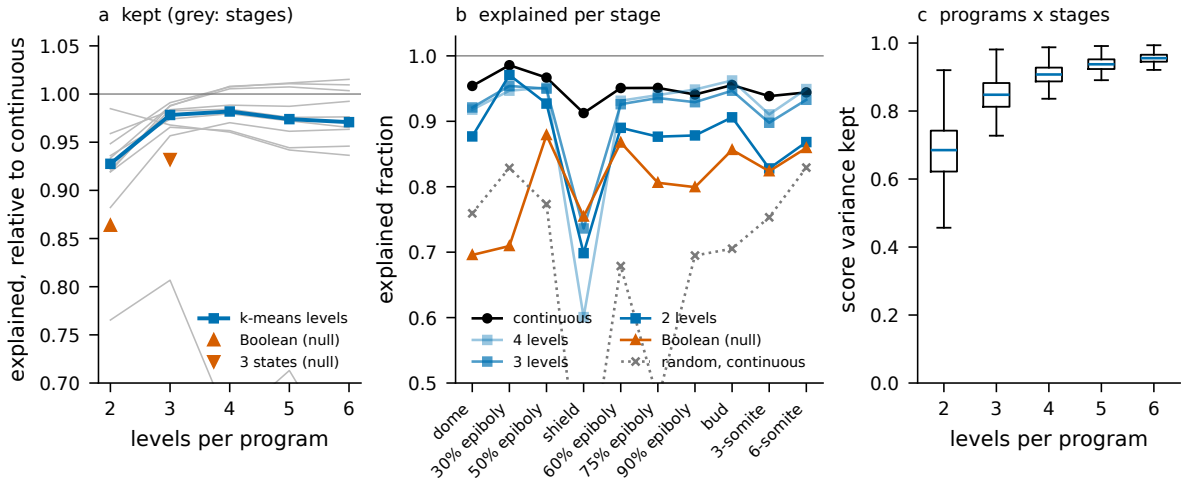


Figure 12: Discretised curated programs (neighbourhood target, heterogeneous evaluation genes). (a) Explained fraction relative to the continuous scores, against the number of k-means levels per program (grey: stages, dome to 6-somite; blue: median), and for the Boolean and 3-state calls from the permutation null (orange; median). (b) Explained fraction per stage (0 = depth and library, 1 = best model). (c) Share of the variance of a program's smoothed score kept by its levels, over all programs and stages.

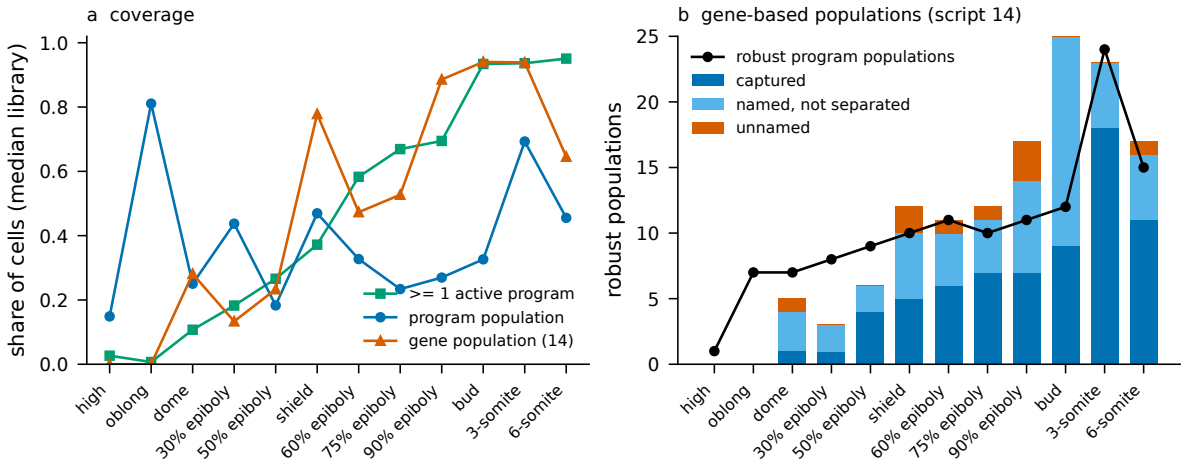


Figure 13: (a) Share of cells (median library) with at least one active curated program, in a robust program-based population, and in a robust gene-based population. (b) Robust gene-based populations per stage, by whether a robust program-based population captures them, a curated program names them without separating them, or no curated program names them; black: robust program-based populations.

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A Supplementary figures and tables

Table S1: Cell filtering per stage. Gene and UMI ranges are the per-stage thresholds of the original study (UMIs including mitochondrial RNA). Cells (GEO) is the number of cells in the GEO matrix; cells and embryos (Table S1) are taken from the supplementary Table S1.

Stage	Genes	UMIs	Cells (GEO)	Cells (Table S1)	Embryos (Table S1)
high	1,000–7,500	1,500–40,000	311	311	56
oblong	625–7,500	1,500–30,000	200	199	76
dome	800–3,800	2,000–20,000	1,174	1,147	28
30% epiboly	625–3,000	1,000–17,500	1,506	1,454	59
50% epiboly	600–4,000	1,500–25,000	5,864	5,695	91
shield	600–2,500	1,000–15,000	1,046	1,020	22
60% epiboly	600–3,500	1,500–22,500	4,201	4,099	79
75% epiboly	600–3,200	1,400–20,000	6,324	6,170	73
90% epiboly	500–3,500	1,000–20,000	5,626	5,474	66
bud	500–3,200	1,000–17,500	7,162	7,112	85
3-somite	500–3,000	1,000–12,500	1,631	1,602	19
6-somite	500–3,000	1,000–12,500	4,460	4,388	40
total			39,505	38,671	694

Table S2: The three tiers of heterogeneous genes, in the libraries of more than 1,000 cells (from dome on for the floors). Floor: minimum residual (median over libraries, and as a factor of the typical φ); the cutoff of a gene is the larger of the floor and the chance level at its mean. Sampling noise: genes selected in an independent simulated library in which no gene is heterogeneous. Well-expressed: genes of the tier with at least 1,000 UMIs; intrinsic noise: the estimated share of them that are merely noisier than typical (FDR of the empirical null).

Tier	Floor	Genes / library	Sampling noise	Well-expressed	Intrinsic noise
weak	0.49 (3.1×)	421–1,277	3–15 (median 8)	44–301	$\leq 50\%$
moderate	0.78 (6.0×)	258–920	1–9 (median 5)	20–183	$\leq 10\%$
strong	1.06 (11×)	157–667	0–5 (median 2)	11–122	$\leq 1\%$

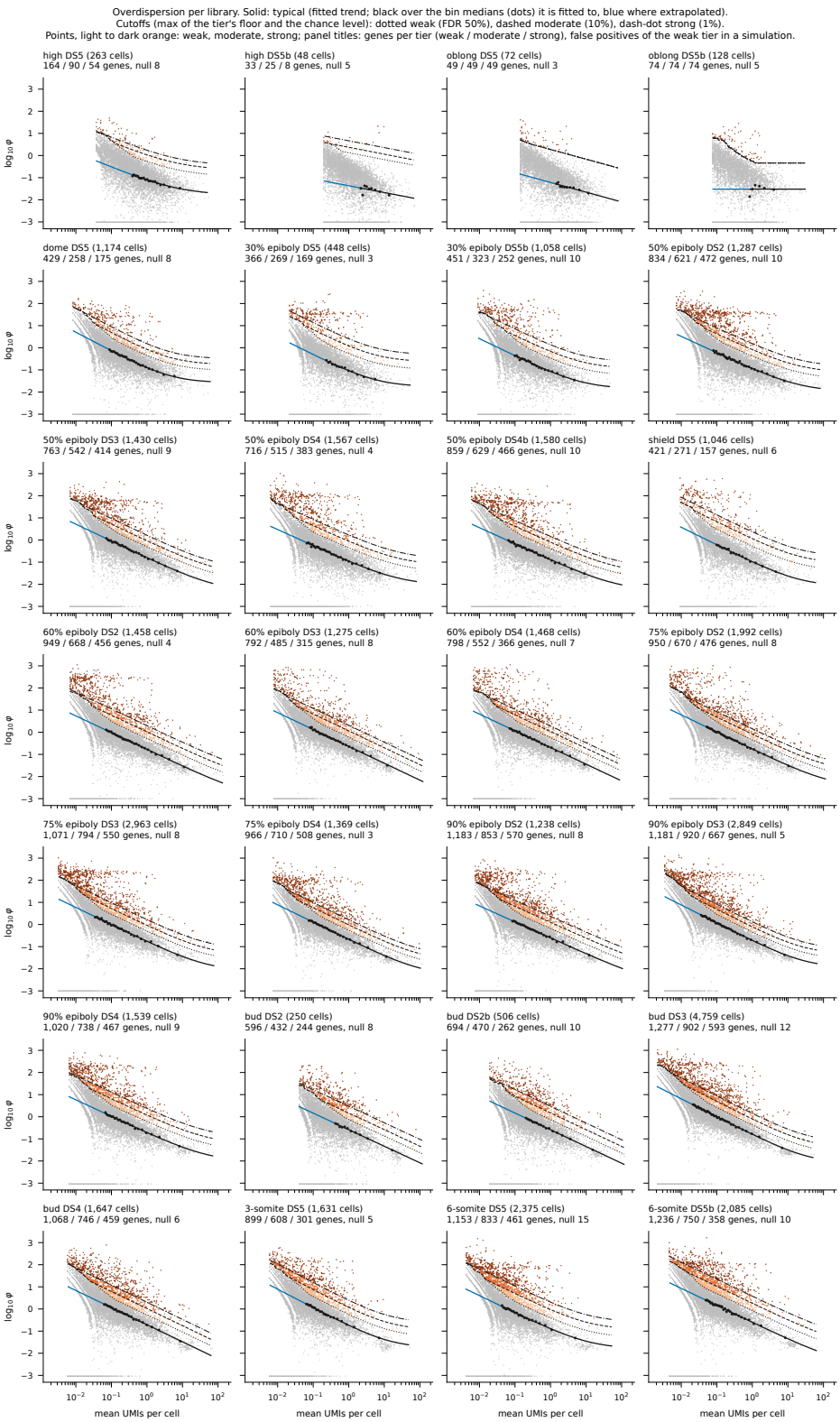


Figure S1: Overdispersion against the mean in every library, as in Figure 5a: estimates, the fitted typical value (blue where extrapolated) and the cutoffs of the three tiers (`scripts/01b_heterogeneous_genes.py`).

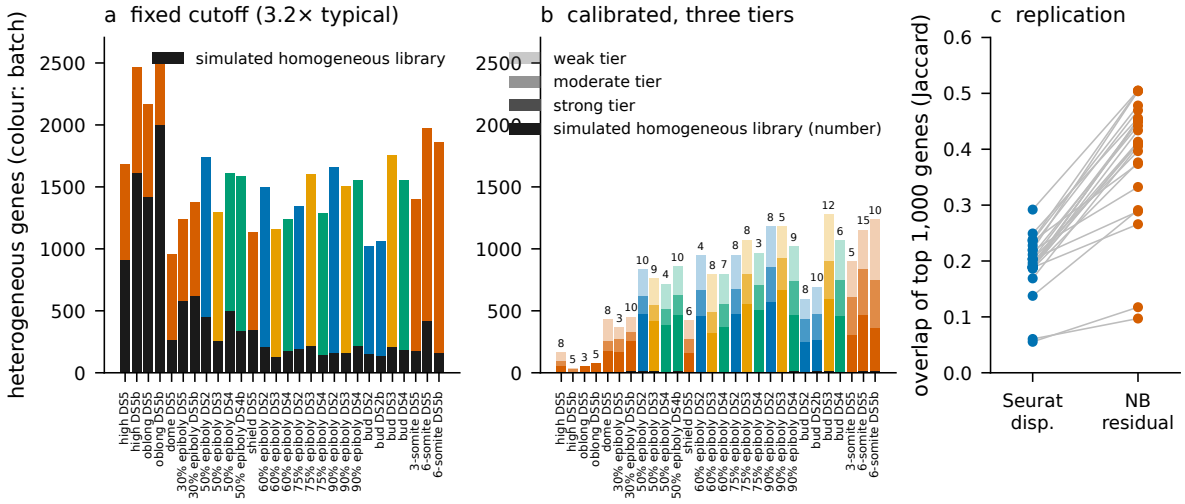


Figure S2: Heterogeneous genes per library. (a) With a fixed cutoff (residual at least 0.5), in the data (coloured by sequencing batch) and in a simulated library with the same depths and means in which no gene is heterogeneous (black, in front). (b) The same for the three tiers (light to full colour: weak, moderate, strong); the simulation is independent of the ones used for the calibration, and its weak-tier genes (at most 15, too few to be visible as bars) are given as numbers. (c) Overlap (Jaccard index) of the top 1,000 genes of two libraries of the same stage, ranked by Seurat's normalized dispersion or by the NB residual; one line per pair of libraries.

Table S3: Heterogeneous genes vs. HVG methods, ranges over the stages from dome on. Top 2,000: genes also heterogeneous, and genes with fewer than 10 UMIs in every library. ZFIN, top k: share of the ZFIN genes with a restricted (Restr.) or whole-organism (Whole) pattern at the stage among the top k genes (k = number of heterogeneous genes). Jaccard: overlap of the top 1,000 genes of two libraries of a stage (median pair per stage, from 30% epiboly on).

	Top 2,000		ZFIN, top k		
Method	Het.	Unmeas.	Restr. %	Whole %	Jaccard
Seurat (v1/v2)	339–917	816–1074	16–40	1–2	0.14–0.29
Seurat (v1/v2), ≥ 3 cells	371–1027	510–849	19–41	1–3	0.16–0.34
Seurat v3 (vst)	420–1296	379–703	26–55	1–3	0.29–0.49
Seurat v3, per library	420–1255	282–755	26–54	1–3	–
Pearson residuals	412–1217	448–792	25–54	1–4	0.24–0.46
Pearson res., per library	412–1203	1–623	25–56	1–5	–
NB residual (01b)	421–1369	0–0	29–59	1–4	0.29–0.50
heterogeneous (01b)	–	–	32–61	1–4	–

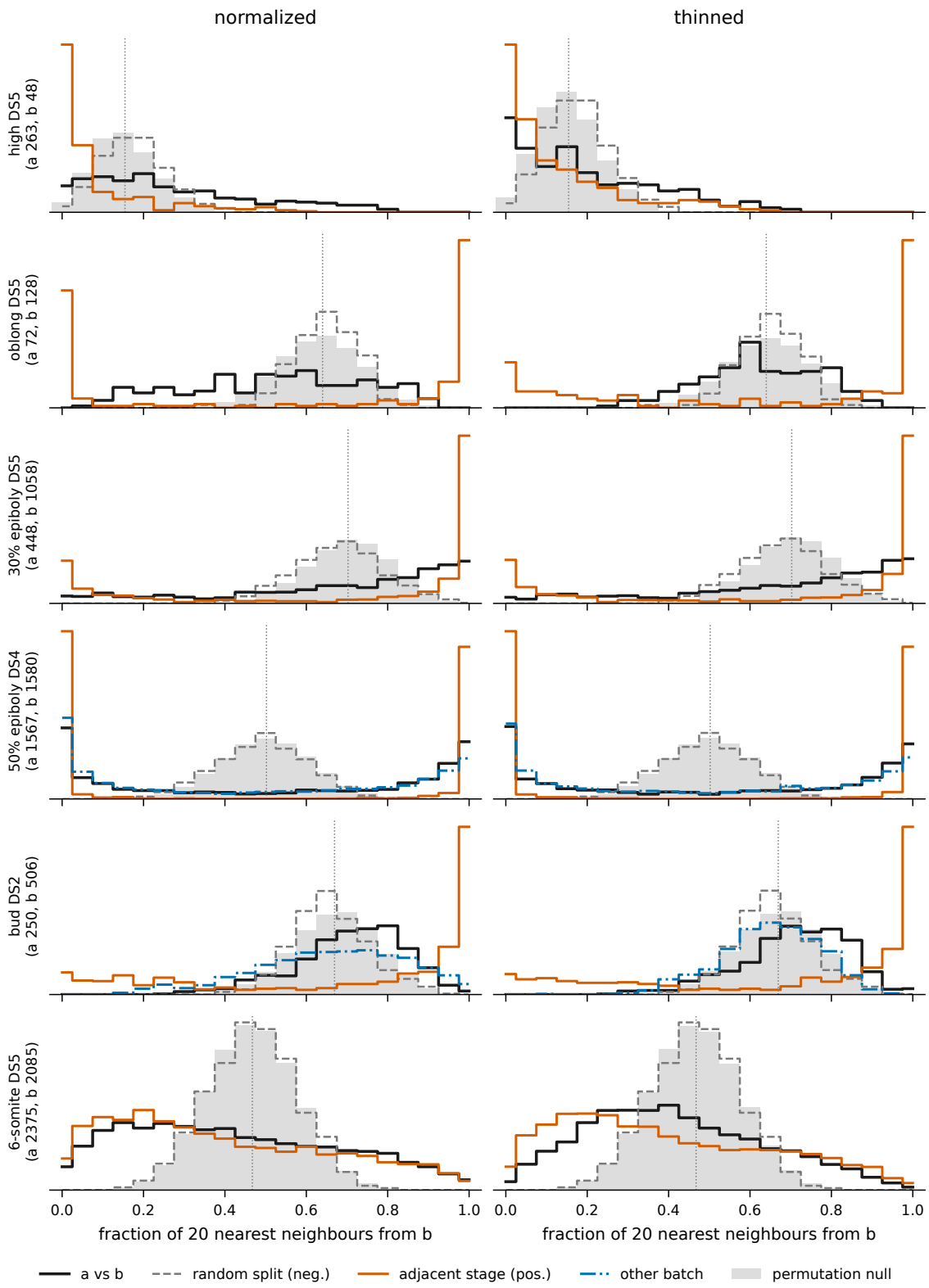


Figure S3: Mixing of the a/b libraries: per-cell distributions of the neighbourhood composition for every comparison of Figure 7 (`scripts/02_replicate_mixing.py`).

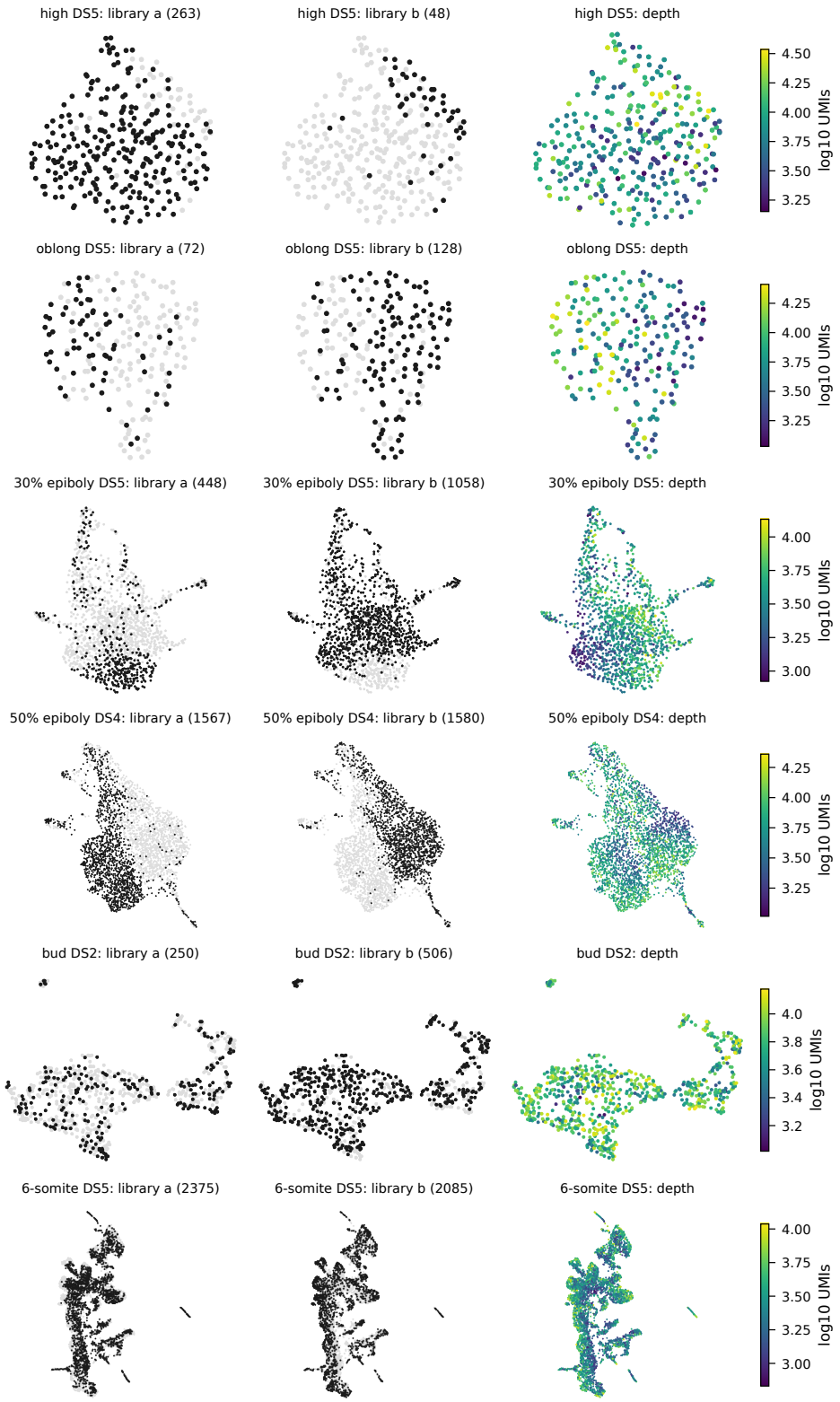


Figure S4: UMAPs of the six a/b pairs (library-size normalization only), coloured by library (`scripts/02_replicate_mixing.py`).

Table S4: Selected genes supporting the explanations of the a/b differences. Mean expression is in UMIs per 10,000 UMIs; the fold change is b relative to a. Detection is the fraction of cells with at least one UMI, after thinning the deeper library to the median depth of the other.

Pair	Gene	Reading	Mean/10k (a/b)		$\log_2$FC	Detected (a/b)	
high DS5	*gbgt1l4*	on/off	0.1	2.3	+3.5	6%	71%
	si:dkey-261j4.3	on/off	0.9	0.1	−2.7	43%	4%
	nanos3	germ plasm	2.3	9.1	+1.9	50%	44%
	dnd1	germ plasm	2.5	6.2	+1.2	43%	21%
	tbx16	early zygotic	2.5	0.5	−2.2	64%	15%
oblong DS5	*aldh1l1*	near-absent in b	0.7	0.1	−2.3	19%	2%
	chia.6	near-absent in b	0.7	0.0	−2.7	21%	1%
	krt8	EVL	0.0	1.1	+3.0	4%	12%
	krt18	EVL	0.5	3.3	+2.5	7%	21%
	cldne	EVL	0.3	2.8	+3.0	4%	18%
50% epiboly DS4	*hsp90aa1.2*	heat shock	8.4	0.4	−4.0	81%	14%
	hsp70.3	heat shock	1.4	0.1	−3.1	34%	3%
	serpinh1b	heat shock	4.4	0.5	−2.9	57%	14%
	si:dkey-108k21.10	on/off	1.8	0.1	−3.2	49%	5%
bud DS2	*myod1*	muscle (adaxial)	0.7	0.1	−2.2	4%	1%
	myl10	muscle (adaxial)	1.7	0.4	−2.0	12%	8%
	actc1a	muscle (adaxial)	0.7	0.2	−1.6	13%	5%
	sox19b	early neuroectoderm	0.2	0.5	+1.0	10%	21%

Stage	Pop.	Cells	Top genes
dome	c7	39	*zgc:110425, zgc:153405, si:ch211-113a14.12, si:ch211-113a14.18, si:dkey-108k21.10, si:dkey-261m9.12*
shield	c10	35	*si:ch211-113a14.18, zgc:153405, si:dkey-108k21.10, zgc:110425, si:ch211-113a14.24, si:dkey-261m9.12*
shield	c16	14	*si:ch211-260e23.9, sesn3, bcl2l14, gadd45aa, mat2al, zgc:64022*
60% epiboly	c11	152	*dlx3b, foxi1, apoc1, apoeb, tfap2a, ved*
75% epiboly	c2	904	*dlx3b, apoc1, apoeb, ved, foxi1, marcksl1b*
90% epiboly	c0	1016	*dlx3b, apoeb, apoc1, marcksl1b, gata3, foxi1*
90% epiboly	c11	238	*cxcl12a, osr1, id3, si:dkey-261j4.5, gata5, apoc1*
90% epiboly	c15	205	*twist1a, cxcl12b, ripply1, cxcl12a, fsta, aplnra*
6-somite	c14	134	*apoeb, apoc1, crabp2a, pfn1, cdx4, cldni*

Table S5: Robust gene-based populations that no curated program names: candidates for missing programs.

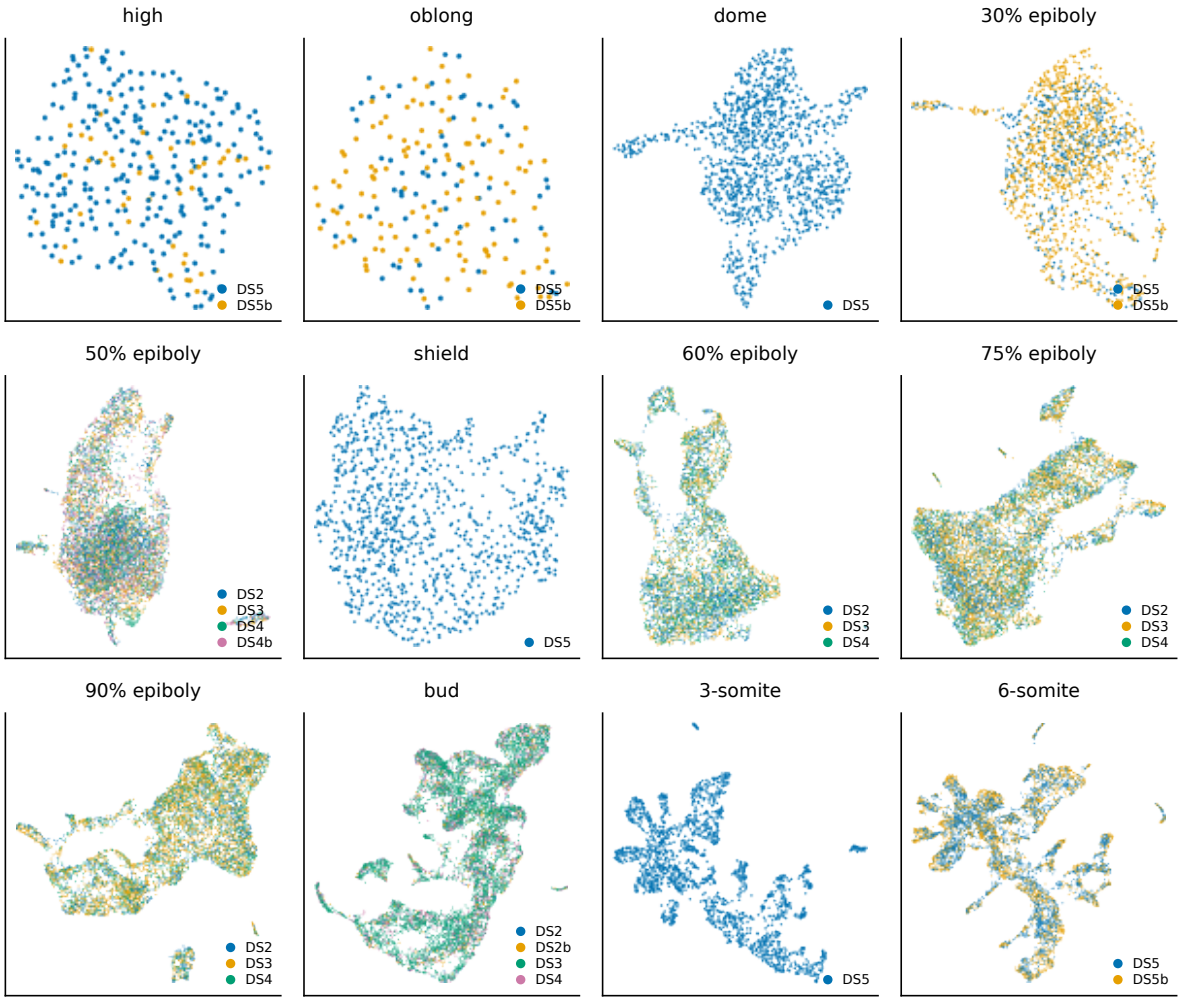


Figure S5: UMAPs of the aligned libraries of every stage; colour: library (sequencing batch, *b* for the second library of a batch).

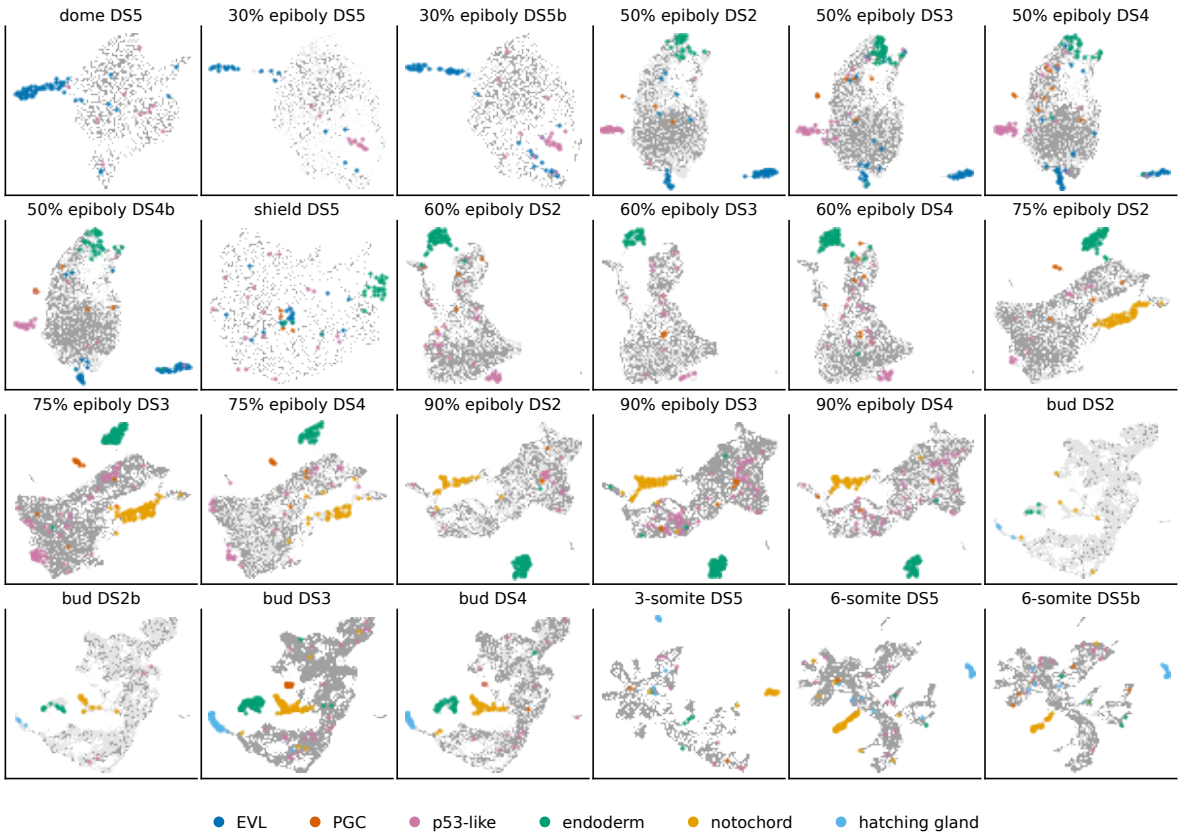


Figure S6: Marker-called populations (check 4) in every library from dome on, on the aligned UMAP of the stage (Figure S5). Dark grey: the cells of the library; light grey: the other libraries of the stage. A population is shown only at the stages where its markers are unambiguous; the marker genes are excluded from the features.

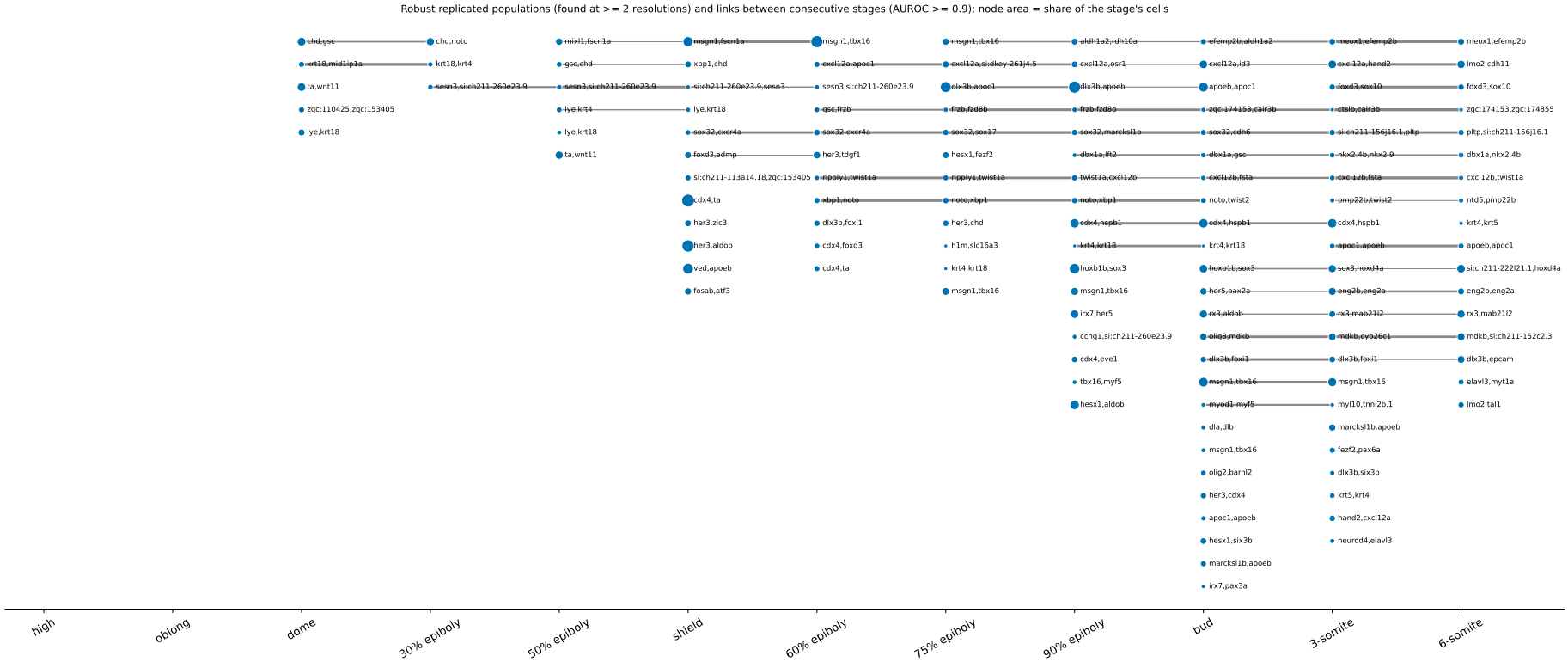


Figure S7: Robust gene-based populations (labelled by their top two genes) linked across consecutive stages into tracks; node area: share of the stage's cells (`scripts/14_replicated_populations.py`).